Chemistry of Carbohydrates

1st year BDS (DIKIOHS)




Each year, 100 metric tons of CO_2 is converted to Carbohydrates by plants

Carbohydrates

- Carbohydrates are the most abundant organic molecules in nature.
- Hydrates of carbon
- Carbohydrates are defined as polyhydroxyaldehydes or ketones or compounds which produce them on hydrolysis.
- Composed of carbon, hydrogen and oxygen

- General molecular formula $C_n (H_2O)_n$
- Some carbohydrates contain Sulphur, Nitrogen or Phosphorus
- Exceptions are acetic acid $C_2H_4O_2$ and lactic acid $C_3H_6O_3$.

Functions of Carbohydrates

- Main sources of ENERGY in body (4kcal/g)
  - RBCs and Brain cells have an absolute requirement of carbohydrates.
- Storage form of energy (starch and glycogen)
- Excess carbohydrate is converted to fat.
- Glycoproteins and glycolipids are components of cell membranes and receptors.
- Structural basis of many organisms.e.g. Cellulose in plants, exoskeleton of insects, cell wall of microbes, mucopolysaccharides and ground substance in higher organisms.

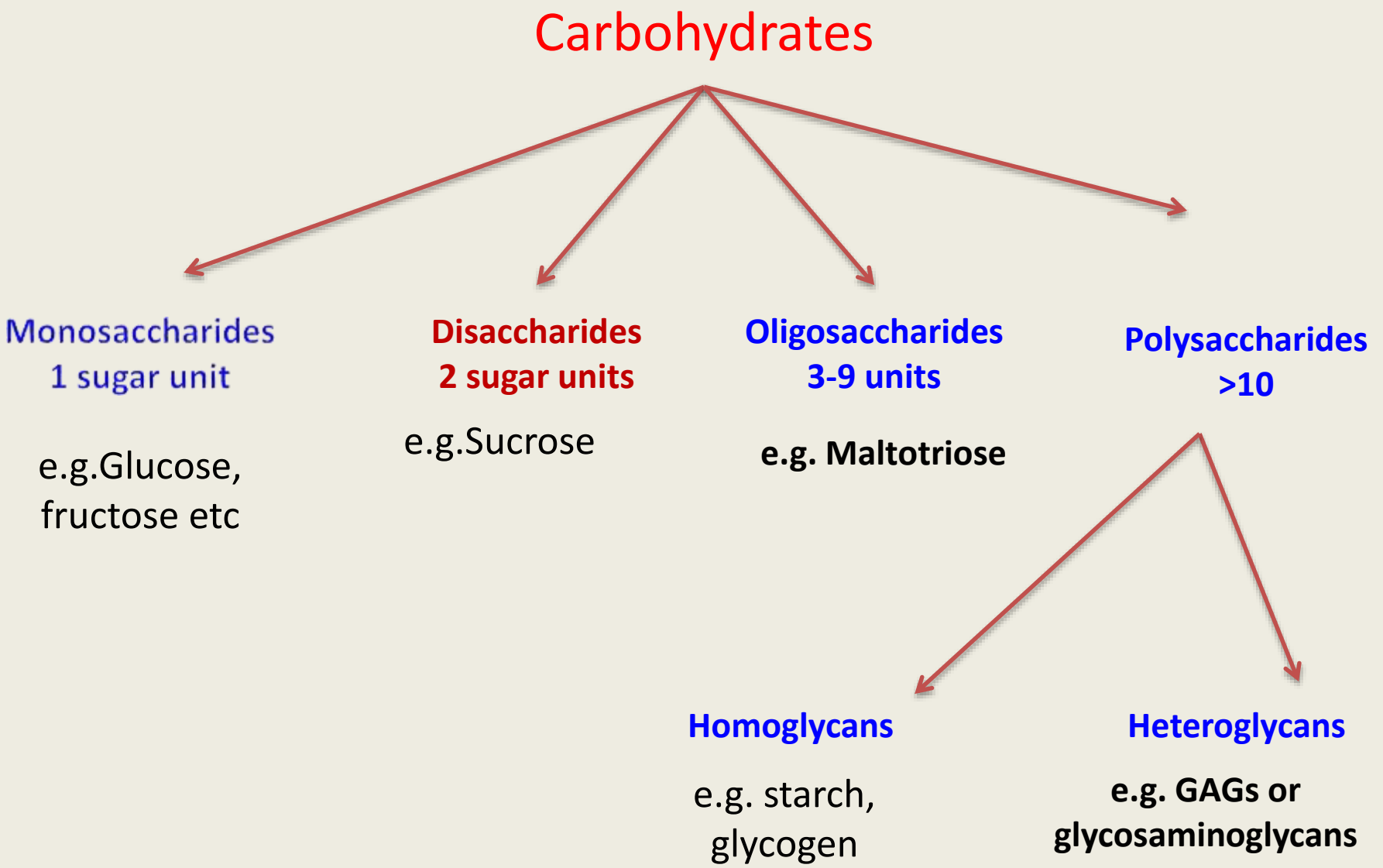
Glycobiology & Sugar Code of Life

- Oligosaccharide Chains Encode Biologic Information
- An enormous number of glycosidic linkages can be generated between sugars.
- For example, three different hexoses may be linked to each other to form over 1000 different trisaccharides.
- Oligosaccharide chains encode **biologic information** and this depends upon their constituent sugars, their sequences, and their linkages.

- The biologic information that sugars contain is expressed via interactions between specific sugars, either free or in glycoconjugates, and proteins (such as lectins) or other molecules.
- **These interactions lead to changes of cellular activity.**

- Thus, deciphering the so-called **‘sugar code of life’** (one of the principal aims of glycomics) entails elucidating all of the interactions that sugars and sugar-containing molecules participate in, and also the results of these interactions on cellular behavior.
- This will not be an easy task, considering the diversity of glycans found in cells.

Classification of Carbohydrates



Monosaccharides

- Molecules having only one actual or potential sugar group

No. of carbon	Generic name	Aldoses	Ketoses
3	Trioses	Aldotriose e.g. glyceraldehyde	Ketotriose e.g. Dihydroxyacetone
4	Tetroses	Aldotetrose e.g. Erythrose	Ketotetrose e.g. Erythrulose
5	Pentoses	Aldopentoses e.g Arabinose, Xylose, Ribose	Ketopentoses e.g. Xylulose, Ribulose
6	Hexoses	Aldohexose e.g. Glucose, Galactose, Mannose	Ketohexose e.g. Fructose
7	Heptoses	Aldoheptose: Glucoheptose	Ketoheptose e.g Sedoheptulose

Pentoses of Physiological importance

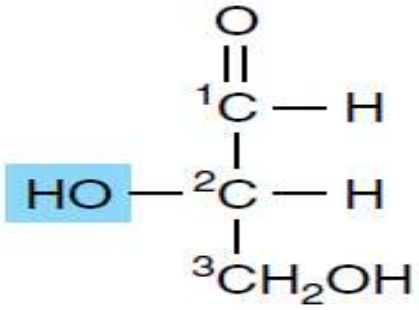
Sugar	Where Found	Biochemical Importance	Clinical Significance
D-Ribose	Nucleic acids.	Structural elements of nucleic acids and coenzymes, eg, ATP, NAD, NADP, flavo-proteins. Ribose phosphates are inter-mediates in pentose phosphate pathway.	
D-Ribulose	Formed in metabolic processes.	Ribulose phosphate is an intermediate in pentose phosphate pathway.	
D-Arabinose	Gum arabic. Plum and cherry gums.	Constituent of glycoproteins.	
D-Xylose	Wood gums, proteoglycans, glycosaminoglycans.	Constituent of glycoproteins.	
D-Lyxose	Heart muscle.	A constituent of a lyxoflavin isolated from human heart muscle.	
L-Xylulose	Intermediate in uronic acid pathway.		Found in urine in essential pentosuria.

Hexoses of Physiological Importance

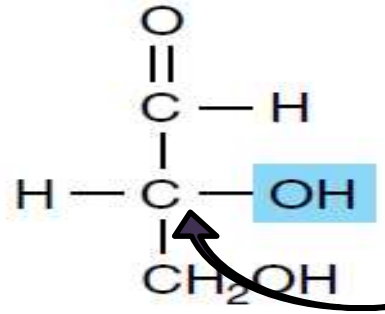
Sugar	Source	Importance	Clinical Significance
D-Glucose	Fruit juices. Hydrolysis of starch, cane sugar, maltose, and lactose.	The "sugar" of the body. The sugar carried by the blood, and the principal one used by the tissues.	Present in the urine (glycosuria) in diabetes mellitus owing to raised blood glucose (hyper-glycemia).
D-Fructose	Fruit juices. Honey. Hydrolysis of cane sugar and of inulin (from the Jerusalem artichoke).	Can be changed to glucose in the liver and so used in the body.	Hereditary fructose intolerance leads to fructose accumulation and hypoglycemia.
D-Galactose	Hydrolysis of lactose.	Can be changed to glucose in the liver and metabolized. Synthesized in the mammary gland to make the lactose of milk. A constituent of glycolipids and glycoproteins.	Failure to metabolize leads to galactosemia and cataract.
D-Mannose	Hydrolysis of plant mannans and gums.	A constituent of many glycoproteins.	

Stereoisomers

- Compounds having *same structural formula, but differing in spatial configuration as known as stereoisomers.*
- Asymmetric carbon:
- Four different groups are attached to the same carbon.
- The reference molecule is glyceraldehyde.
- *All monosaccharides can be considered as molecules derived from glyceraldehyde by successive addition of carbon atoms.*

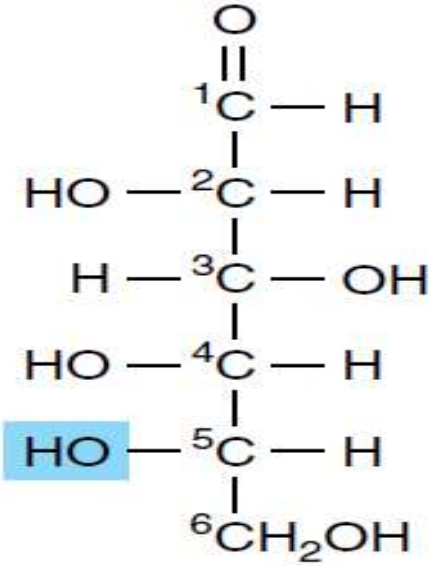


L-Glycerose
(L-glyceraldehyde)

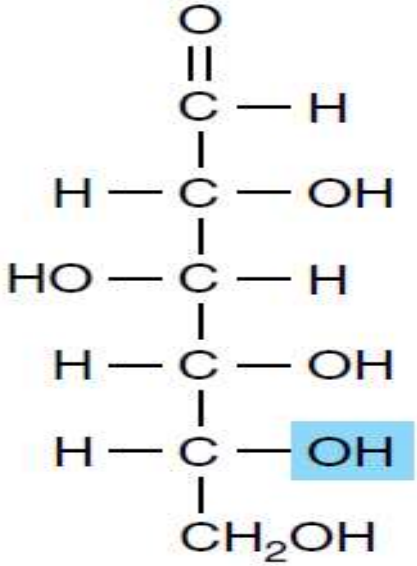


Penultimate Carbon

D-Glycerose
(D-glyceraldehyde)



L-Glucose

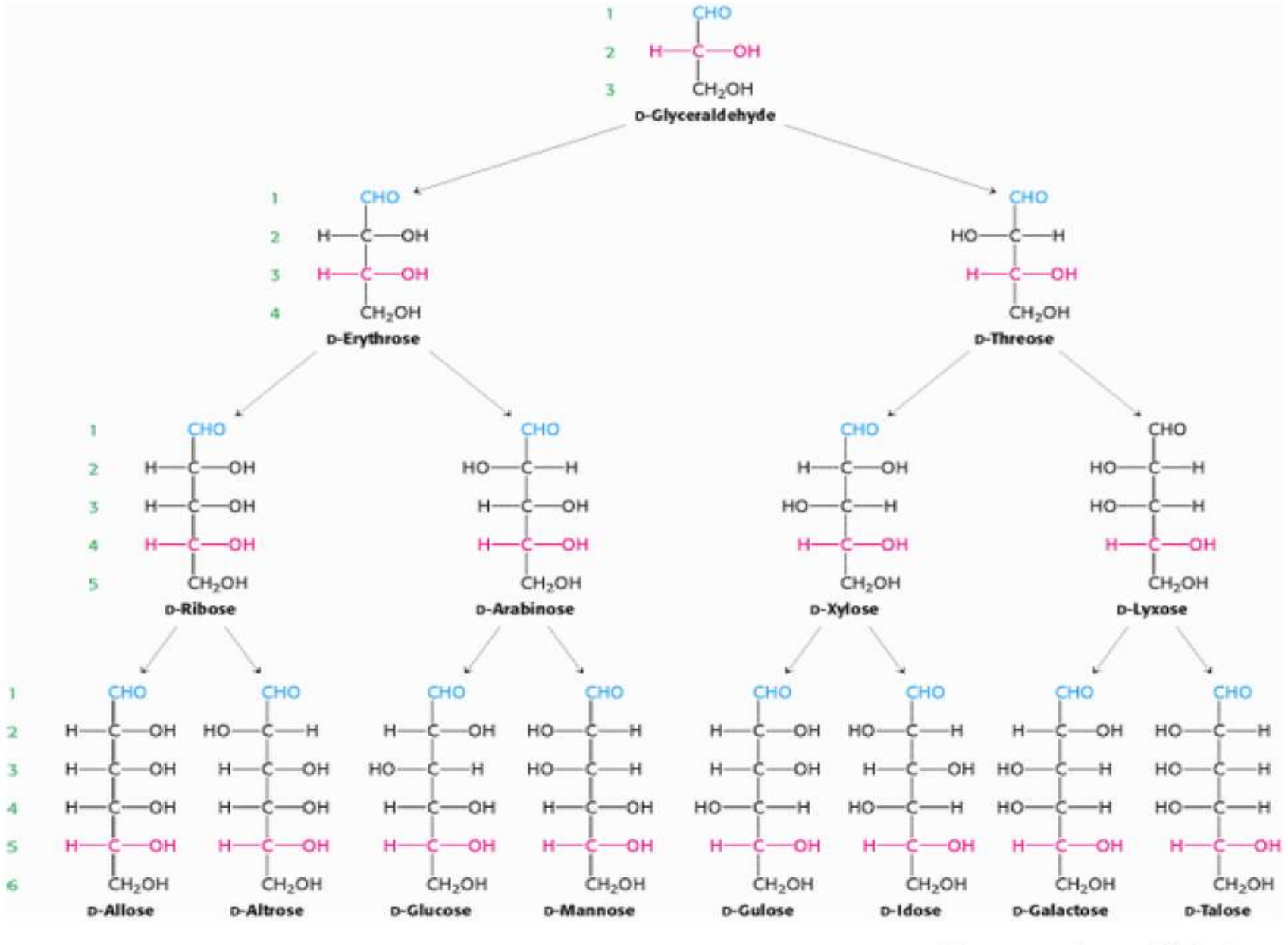


D-Glucose

Stereoisomers

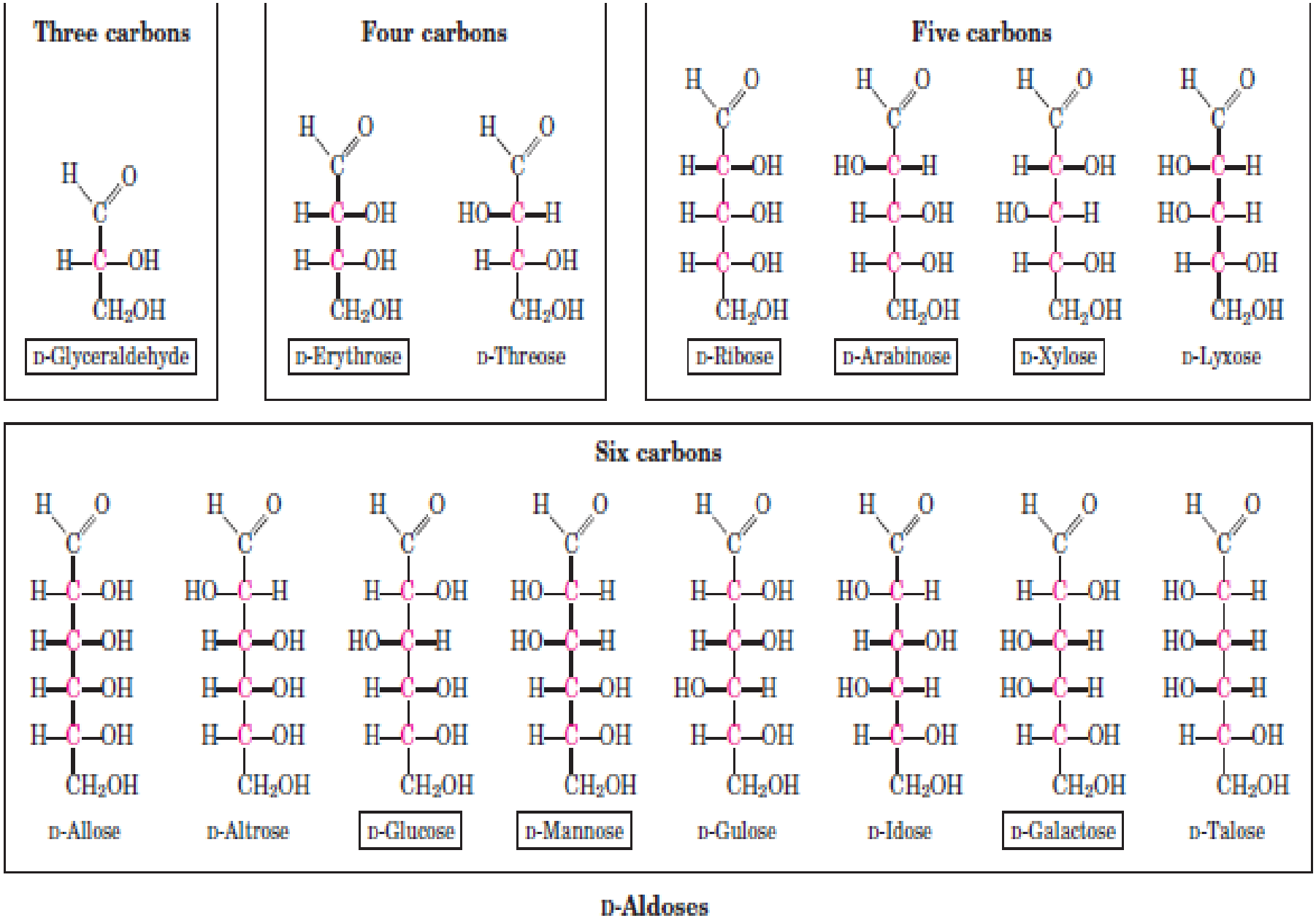
D and L Isomers of glucose

- D and L Isomers are mirror images of each other.
- The *spatial orientation of H & OH groups on the C-atom (C5 for glucose), adjacent to the terminal primary alcohol carbon determines whether the sugar is D or L Isomer.*
- If the OH group is on the right side, the sugar is of D-Isomer.
- If the OH group is on the left side, the sugar is of L-Isomer.
- Mammalian tissues have D- sugars.

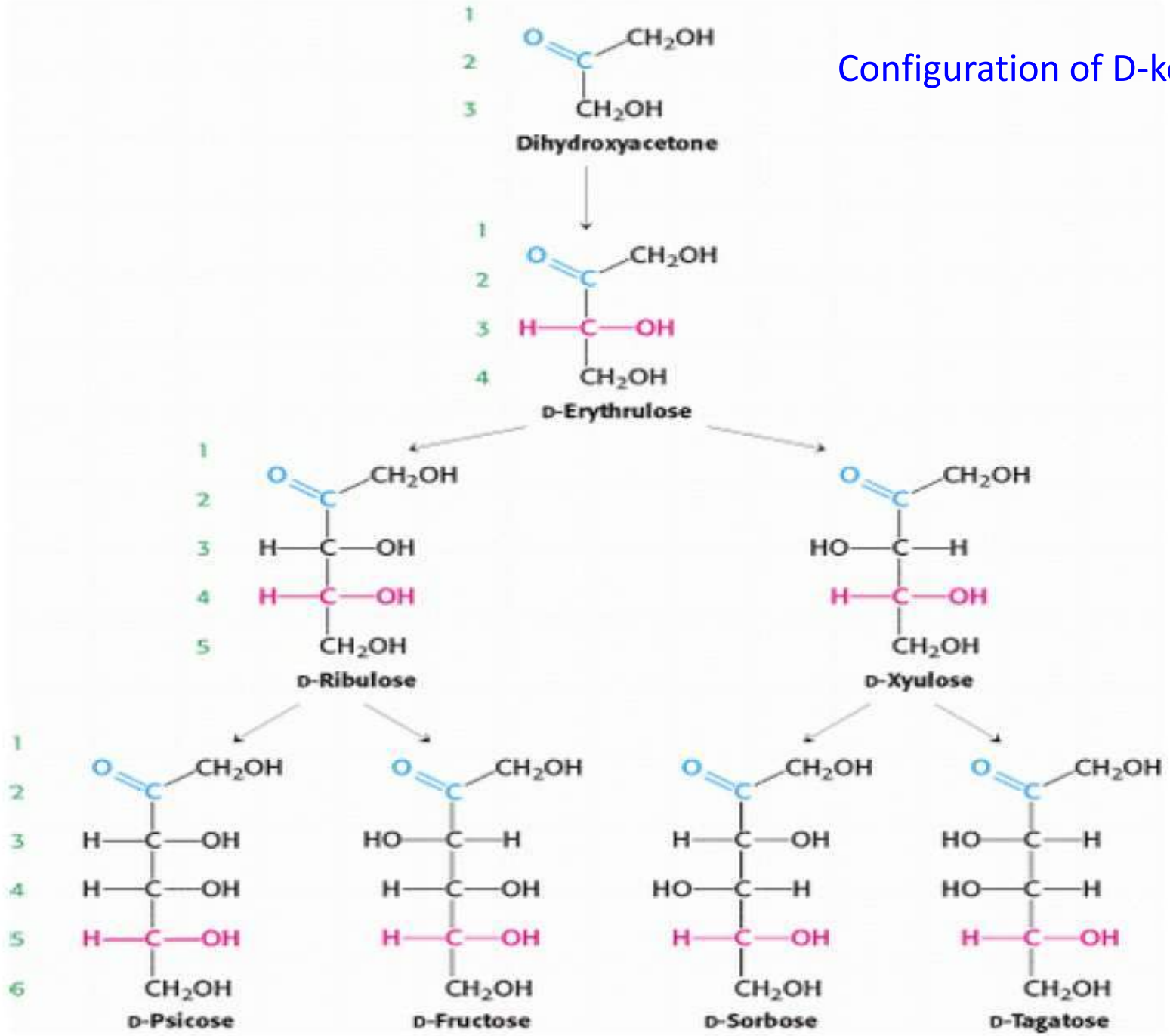


Configuration of D-aldoses

Aldoses

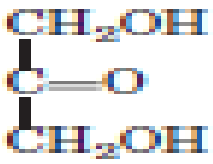

Three carbons
D-Glyceraldehyde
Four carbons
D-Erythrose
D-Threose
Five carbons
D-Ribose
D-Arabinose
D-Xylose
D-Lyxose
Six carbons
D-Allose
D-Altrose
D-Glucose
D-Mannose
D-Gulose
D-Idose
D-Galactose
D-Talose
D-Aldoses

Configuration of D-ketoses



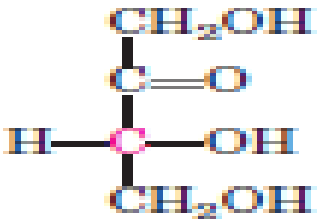
Ketoses

Three carbons



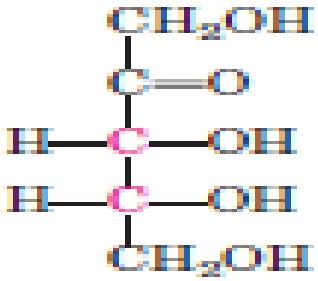
Dihydroxyacetone

Four carbons

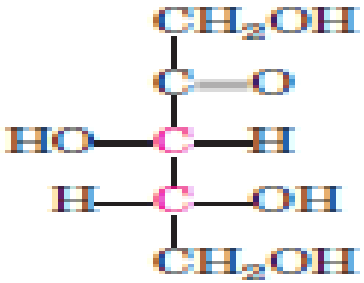


D-Erythrulose

Five carbons

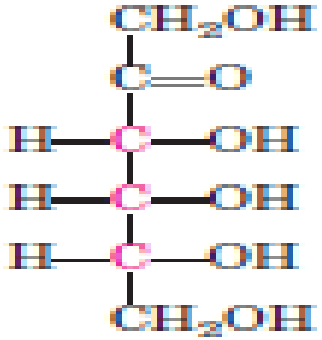


D-Ribulose

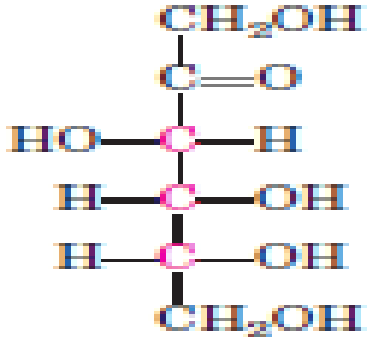


D-Xylulose

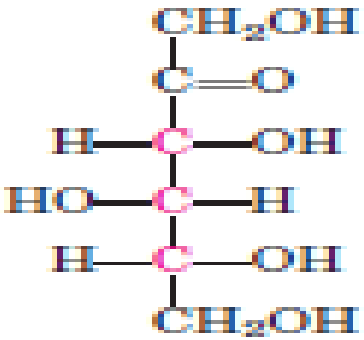
Six carbons



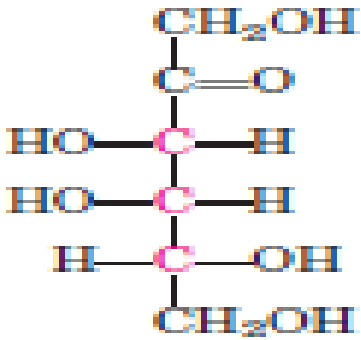
D-Psicose



D-Fructose



D-Sorbose



D-Tagatose

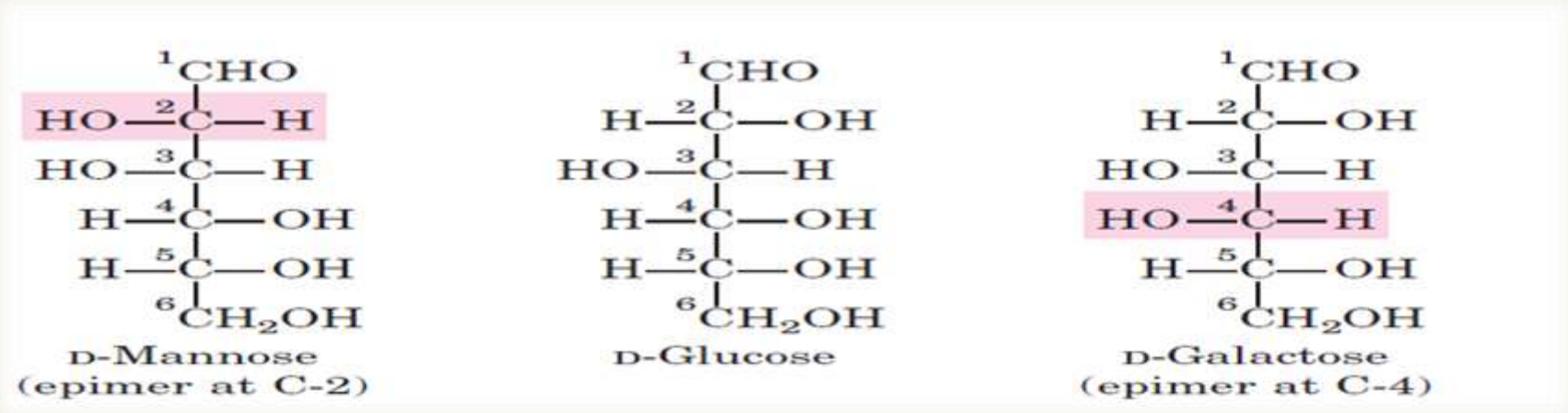
D-Ketoses

Optical activity

- Optical activity is a characteristic feature of compounds with asymmetric carbon atom.
- When a beam of *polarized light is passes through a solution of an optical isomer, it will be rotated either to the right or left.*
- Depending on the rotation, molecules are called dextrorotatory (+) or levorotatory (-).
- Racemic mixture: If D & L isomers are present in equal concentration, it is known as racemic mixture.
- NOTE: Racemic mixture does not exhibit any optical activity.

Epimers

- If two monosaccharides differ from each other in their configuration around a single specific carbon atom, they are referred as epimers to each other.
- Glucose & galactose are C4-epimers
- Glucose & mannose are C2-epimers
- Inter-conversion of epimers is known as epimerization, epimerases catalyzes this reaction.

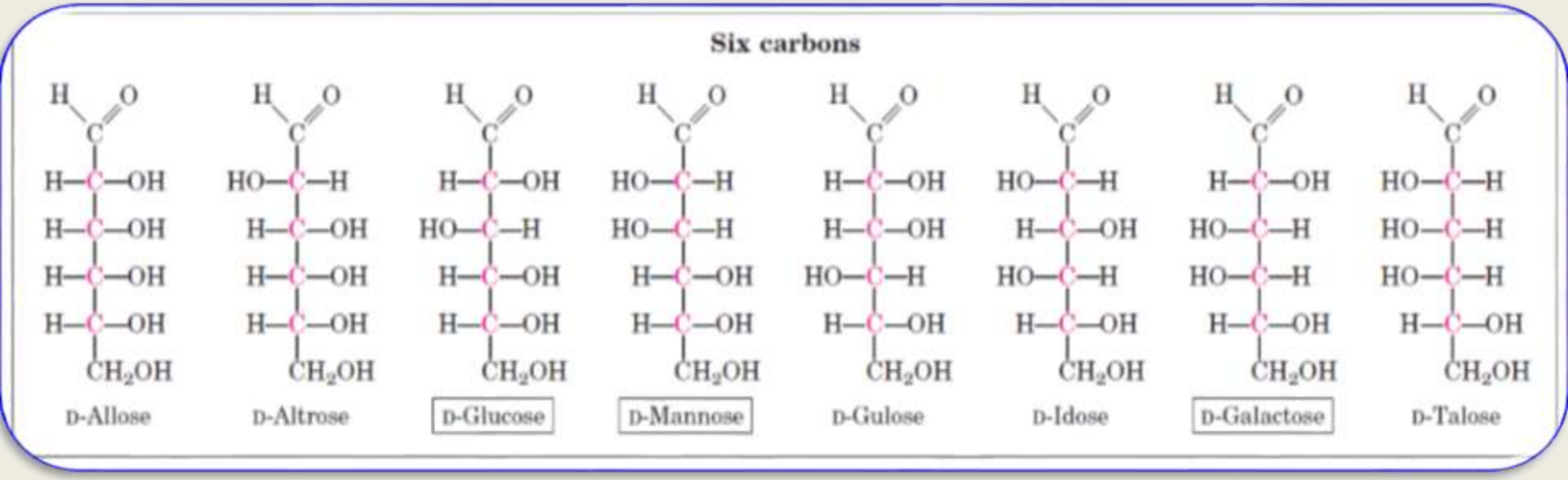


Enantiomers

- Enantiomers are a special type of stereoisomers, that are mirror images of each other.
- Majority of sugars in higher animals are of D-type.

Diastereoisomers

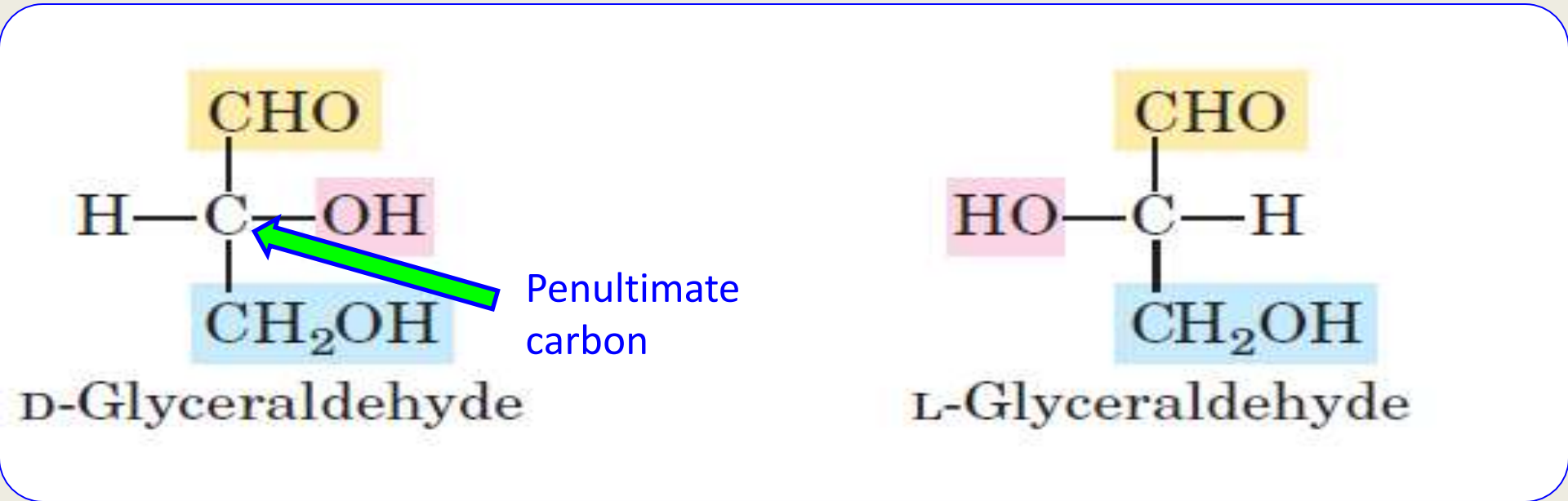
- The term diastereomers is used to represent the stereoisomers that are not mirror images of one another.
- Configurational changes with regard to C2 , C3 and C4 will produce eight different monosaccharides.



Total D + L forms = 16 isomers of glucose

Physical Properties

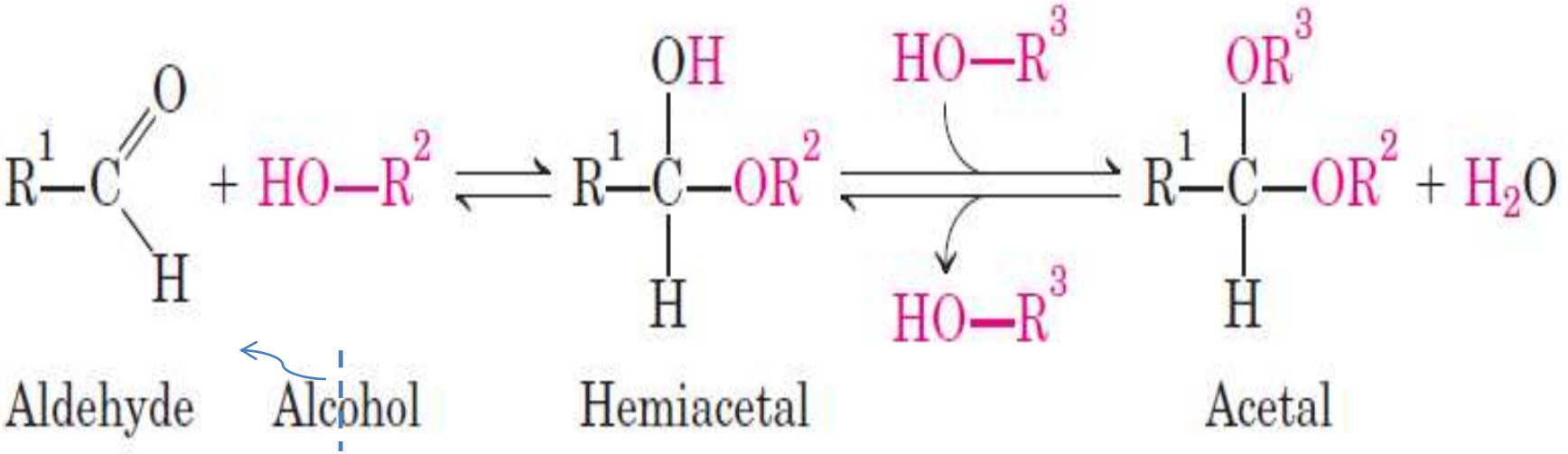
- Reference Carbon atom of Sugars:-
- Penultimate carbon atom is the reference carbon atom for naming sugars.
- All monosaccharides can be considered as molecules derived from glyceraldehyde by successive addition of C units.

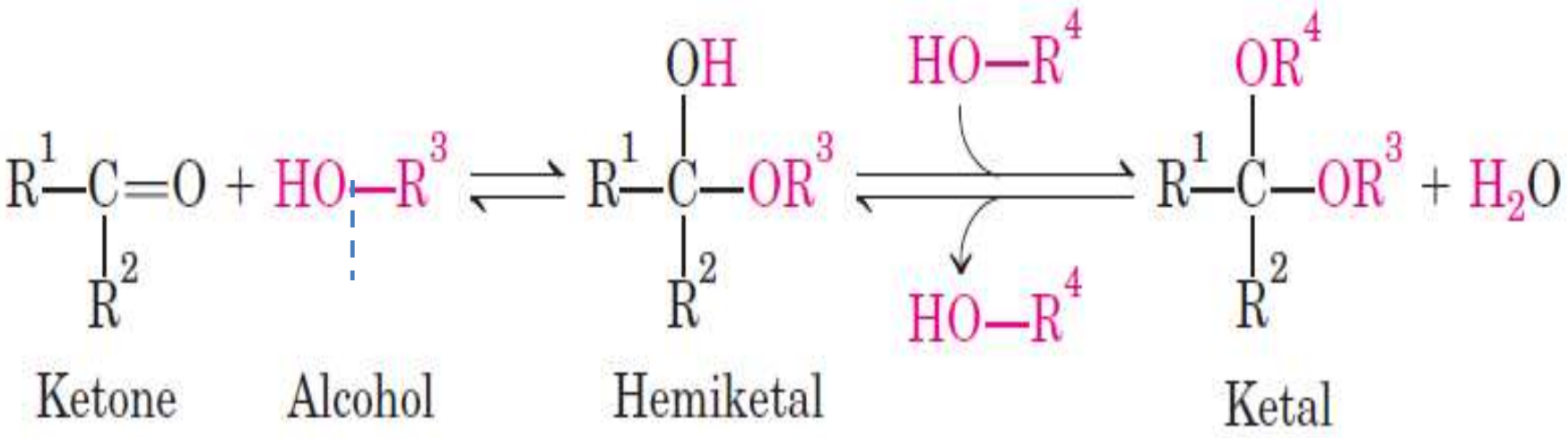


Glycosides

- When the hemi-acetal group (hydroxyl group of the anomeric carbon) of a monosaccharide iscondensed with an alcohol or phenol group, it is called a glycoside.
- The non-carbohydrate group is called aglycone.

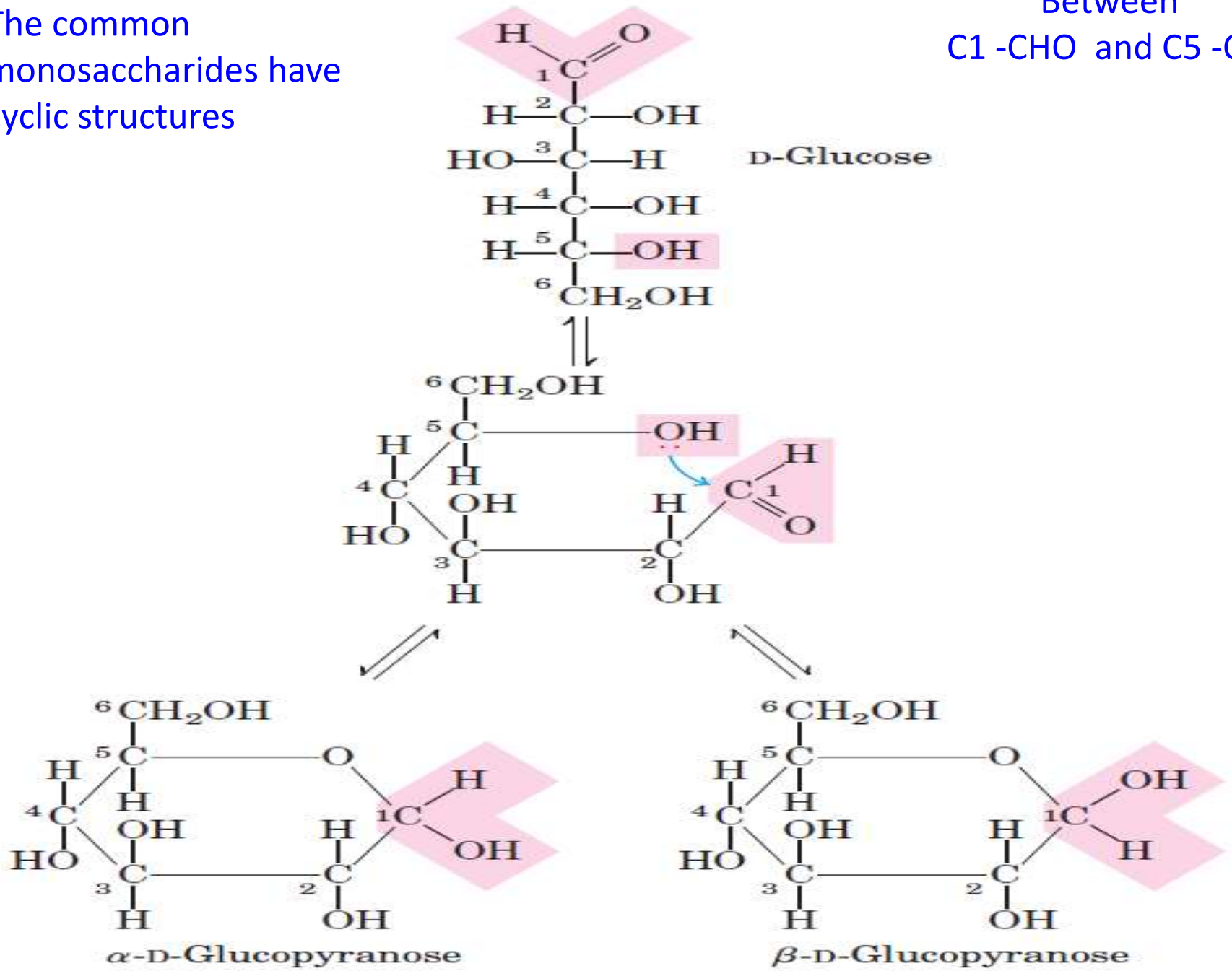
Formation of hemiacetals and hemiketals





The common monosaccharides have cyclic structures

Between
C1 -CHO and C5 -OH



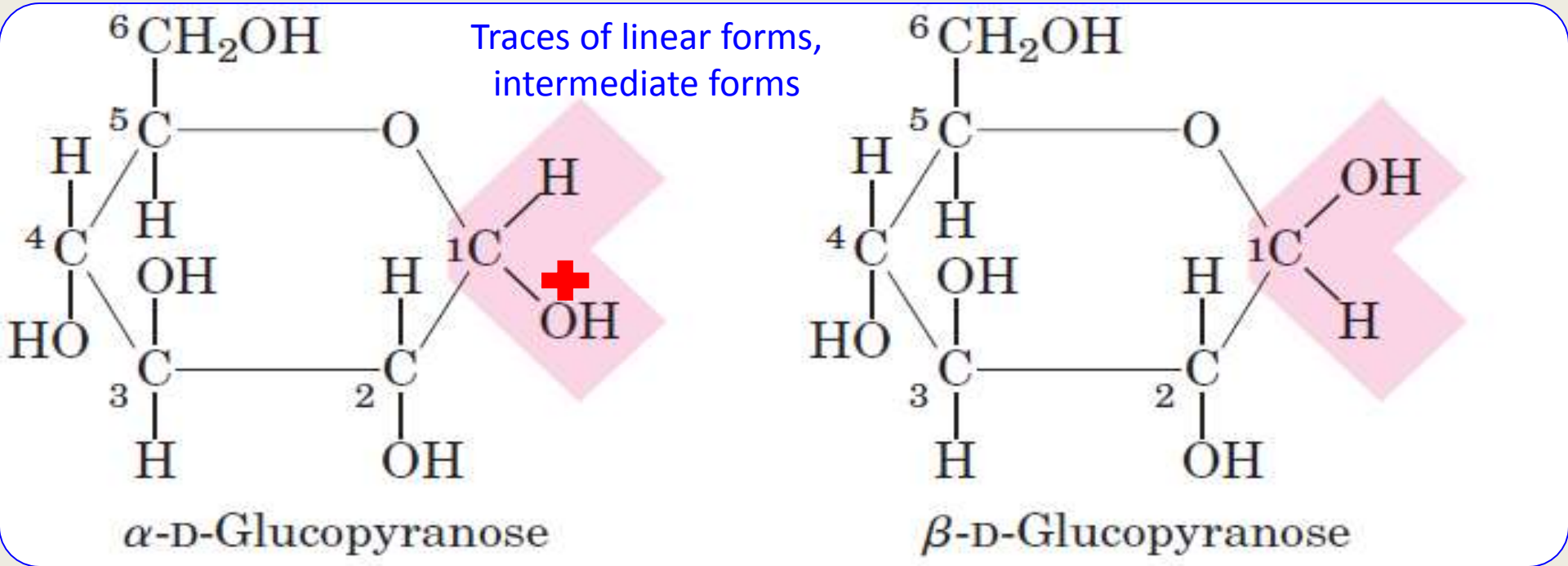
Anomerism

- Anomers have same composition but differ in the orientation of groups around anomeric carbon atom.
- Anomeric carbon is a carbonyl carbon atom, e.g. 1st carbon atom in glucose is anomeric carbon atom.
- Carbonyl carbon atom becomes asymetric because of ring structures of monosaccharides in solution thus anomers are encountered in cyclic structures of monosaccaharides.
- The alpha & beta cyclic forms of D-glucose are known as anomers.
- *They differ from each other in the configuration only around C1 known as anomeric carbon.*
- The hemiacetal (or carbonyl) carbon atom is called the anomeric carbon.
- In case of alpha anomer, the OH group held by anomeric carbon is on the opposite side of the group CH_2OH of sugar ring.

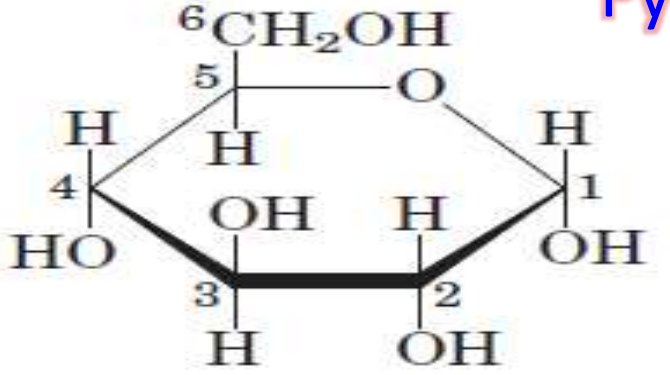
Expression

- Anomers are expressed as α and β forms.
- In α form “OH” group is below the plane (OH group is oriented away from the oxygen atom)
- In β form “OH” group is above the plane (OH group is oriented towards the oxygen atom)

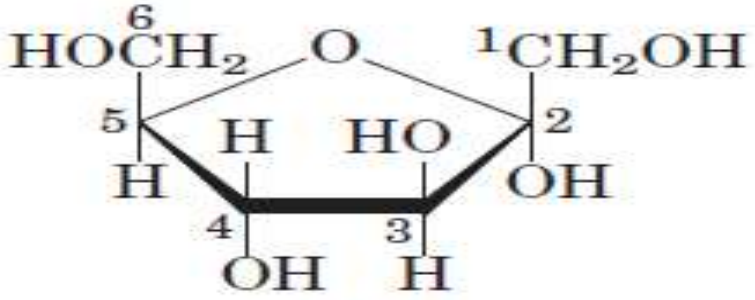
- Mutation:
- When D-glucose is crystallized at room temperature and a fresh solution is prepared, its specific rotation of polarized light is 112°; but after 12- 18 hrs it changes to +52.5 °
- This change in rotation with time is called as mutarotation.
- Glucose has two anomers α and β.



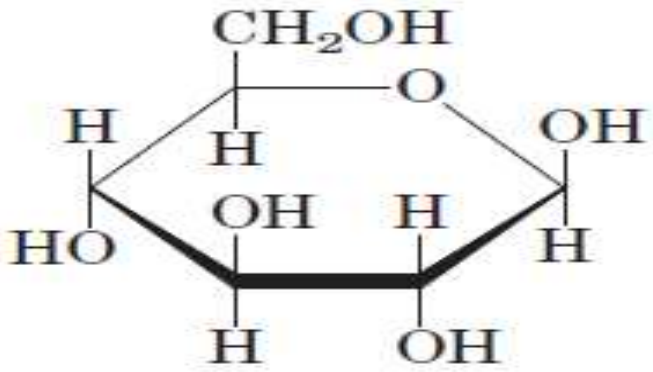
Pyranoses and Furanoses



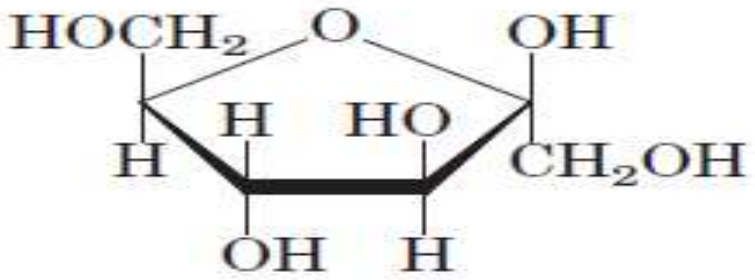
α-D-Glucopyranose



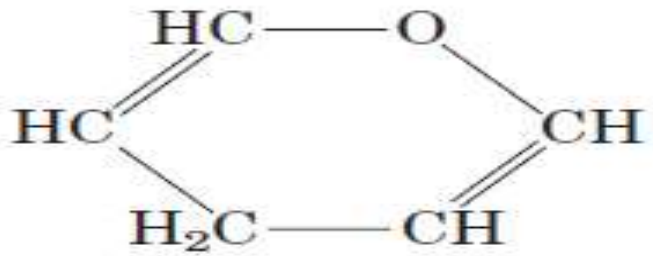
α-D-Fructofuranose



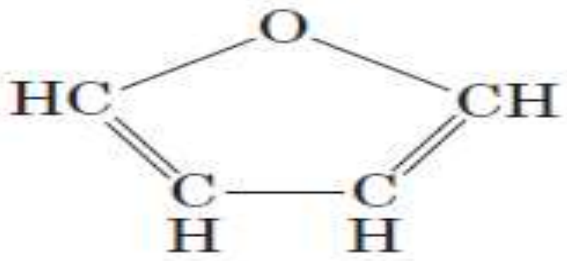
β-D-Glucopyranose



β-D-Fructofuranose

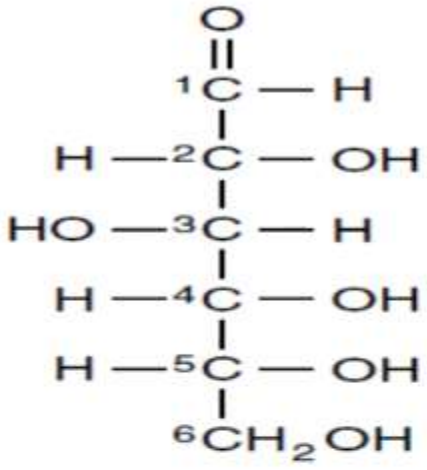


Pyran

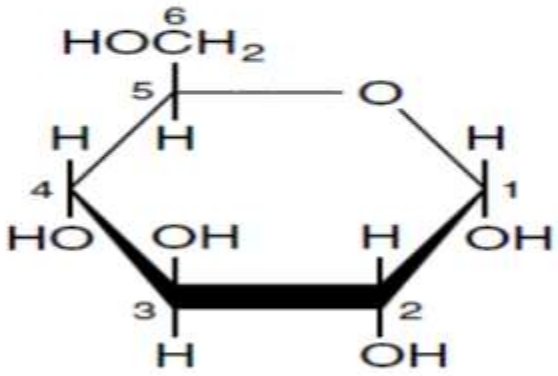


Furan

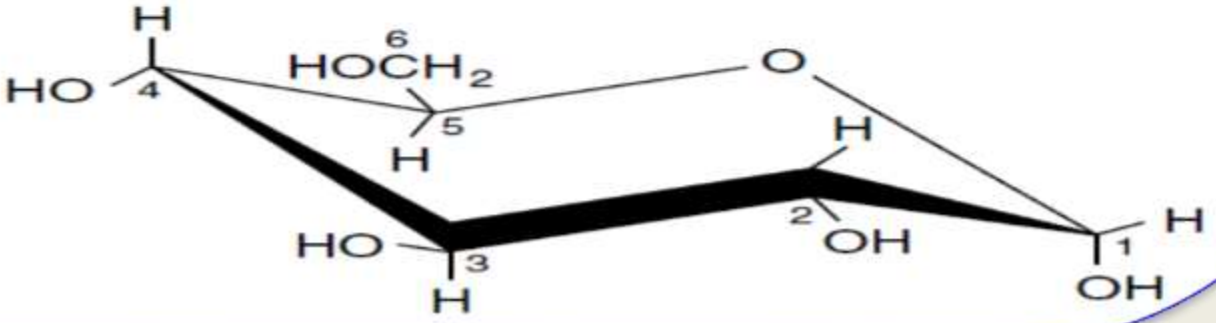
A


O
1C — H
H — 2C — OH
HO — 3C — H
H — 4C — OH
H — 5C — OH
6CH2OH

B


6
HOCH2
5
O
H
H
H
4
1
HO
OH
H
OH
3
2
H
OH

C


H
HO
4
6
HOCH2
5
O
H
H
HO
3
H
2
OH
1
H
OH

Chemical Properties of Carbohydrates

Reactions of monosaccharides

- Tautomerization:
- The process of shifting a hydrogen atom from one carbon atom to another to produce enediols is known as tautomerization.
- Reducing properties:
- In mild alkaline solutions, carbohydrates containing a free sugar group (aldehyde or ketone) will tautomerise to form enediols , where two hydroxyl groups are attached to the double-bonded carbon atoms.
- Since enediols are powerful reducing agents in alkaline medium.
- When oxidizing agents like cupric ions are present , sugars form a mixture of carboxylic acids by breaking at the double bonds.

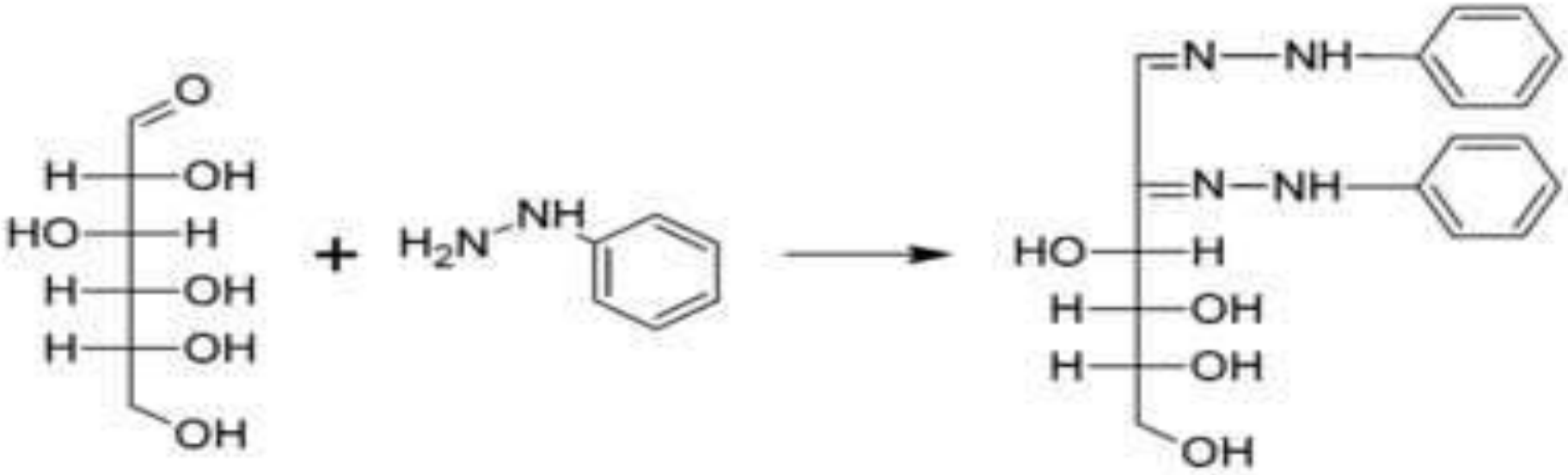
Benedict's test

- Procedure: 0.5 (8 drops) ml urine + 5ml Benedict's reagent & boil for 2 mins.
- Interpretation:
- Drawback – test is not specific for glucose
- Fehling's test: No intermediate colors are formed as because over there a powerful reducing agent KOH is used.

Observation	Inference
No change in colour	No sugar
Green colour	0-0.5mg% +
Yellow	0.5-1.0mg% ++
Orange	1.0-1.5mg% +++
Brick red	1.5-2mg% ++++

Osazone formation

- All reducing sugars will form osazones with excess of phenylhydrazine when kept at boiling temperature.
- Osazones are insoluble.
- Osazones of individual sugars have characteristic crystal from
- The differences in glucose, fructose and mannose are dependent on 1st and 2nd C & this difference is masked when Phenyhydrazine reacts with these two carbons.
- So, Glucose, Fructose and Mannose give broom shaped osazones.



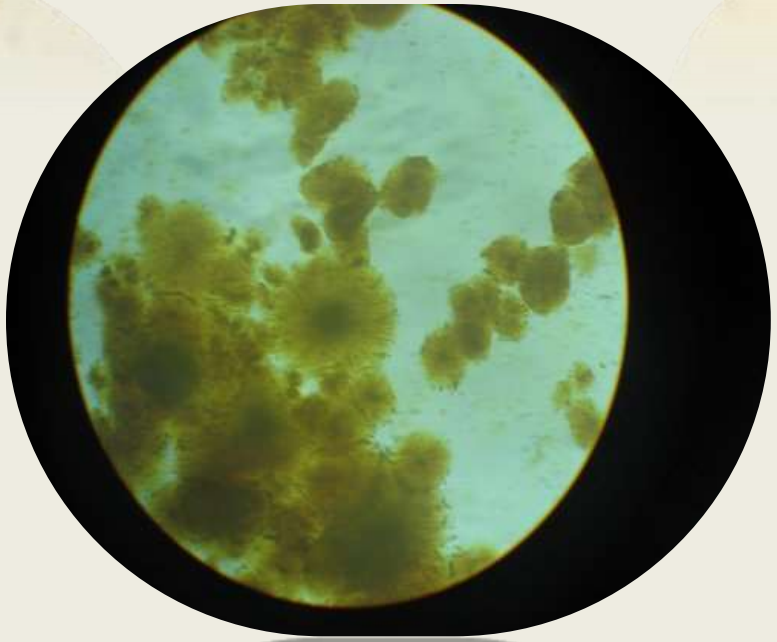
Osazones



Glucosazone
(broom shaped)



Maltosazone
(star shaped)



Lactosazone
(powder puff shaped)

Oxidation of Sugars

1) Mild Oxidation Conditions: e.g. hypobromous acid, *the aldehyde group is oxidized to carboxyl group to produce ALDONIC acids.*

   Glucose → Gluconic acid

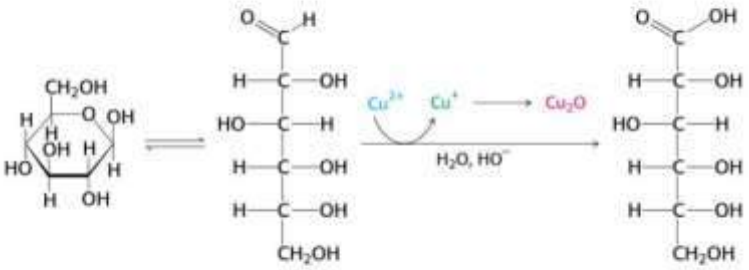
   Mannose → Mannonic acid

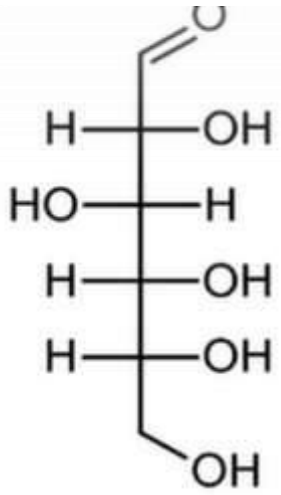
   Galactose → Galactonic acid

2) When Aldehyde group is protected, and *the molecule is oxidized, esp. in the body, the last C is oxidized to COOH producing URONIC acids.*

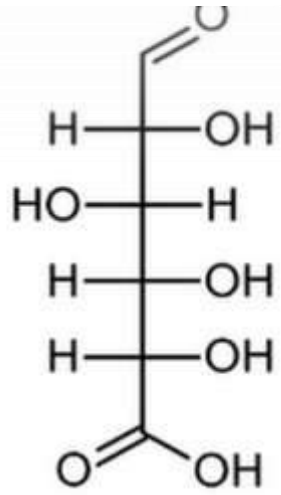
   Imp- Glucuronic acid is used in

   body to synthesize heteropolysaccharides and also for conjugation of various substances.





Glucose



Glucuronic acid

- Under strong oxidation conditions(e.g strong acids-HNO_3)
- *BOTH groups are oxidized to produce dicarboxylic acids called acids called SACCHARIC acids*

Glucose --> Glucosaccharic acid

Mannose --> Mannaric acid

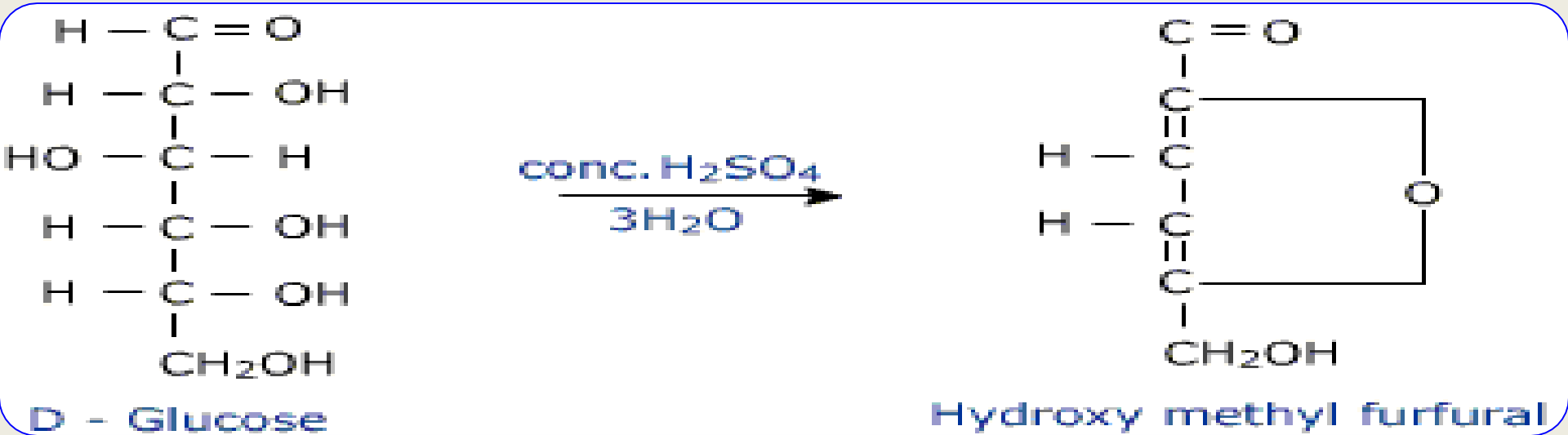
Galactose --> Mucic acid

- Oxidation by glucose oxidase:- GOD-POD method

Furfural formation

- Monosaccharides when treated with concentrated H_2SO_4 undergoes dehydration with the removal of 3 molecules of water.
- Hexoses give hydroxymethyl furfural and pentoses give furfural.
- Furfurals condense with phenolic compounds to give various colors.

  E.g. Molisch's test: General test for carbohydrates (H_2SO_4 and α-naphthol)
- Rapid Furfural and Seliwanoff's test: Tests for presence of keto group



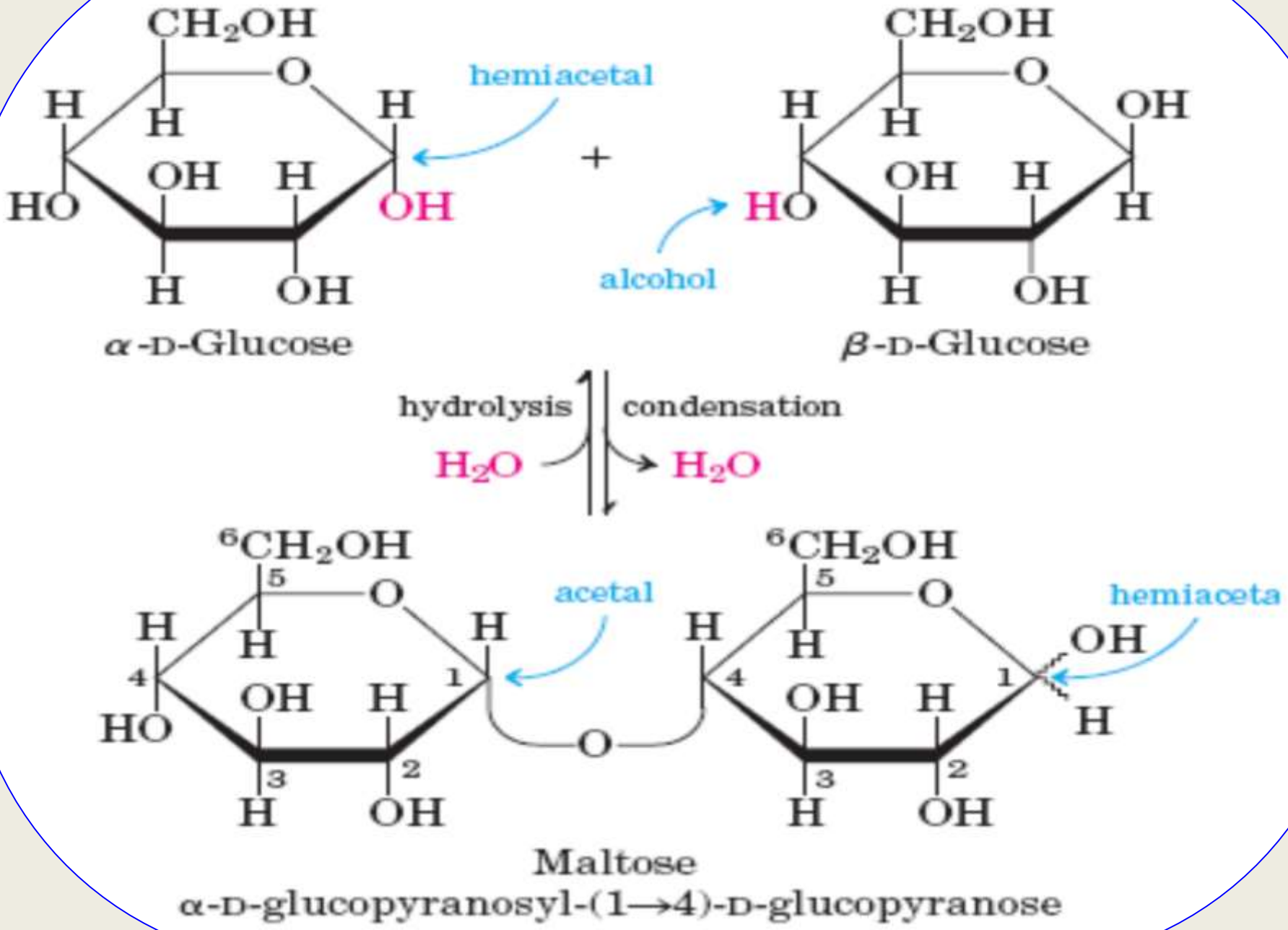
Reduction to form alcohols

- When treated with reducing agents such as sodium amalgam, hydrogen can reduce sugars.
- Aldose yields corresponding alcohol.
- Ketoses form two alcohols because of appearance of new asymmetric carbon in this process.

  D-Glucose → D-Sorbitol ← D-Fructose → D-Mannitol
- Sorbitol and Mannitol are used to identify bacterial colonies.
- Mannitol is used to reduce intracranial pressure by forced diuresis.
- *The osmotic effect of sorbitol and dulcitol produces changes in tissues when they accumulate in abnormal amounts. E.g cataract*

Glycosides

- When the hemi-acetal group (hydroxyl group of the anomeric C) of a monosaccharide is condensed with an alcohol or phenol group, it is called as a glycoside.
- The non-carbohydrate group is called aglycone.
- Glycosides are non –reducing (WHY ?) but they may be hydrolyzed by boiling with dilute acids.
- α- glycosides are hydrolyzed by maltase from yeast, while beta-glycosides are hydrolyzed by Emulsin from almonds.
- So enzyme hydrolysis affords a method to distinguish b/w two forms.


hemiacetal
alcohol
α-D-Glucose
β-D-Glucose
hydrolysis
condensation
H2O
H2O
acetal
hemiaceta
Maltose
α-D-glucopyranosyl-(1→4)-D-glucopyranose

Important Glycosides

Sugar	Aglycon	Glycoside	Source	Importance
Glucose	Phloretin	Phlorizin	Rose bark	Renal damage
Galactose Xylose	Digitogenin	Digitonin	Leaves of foxglove	Cardiac stimulant
Glucose	Indoxyl	Plant indican	Leaves of indigofera	Stain

Formation of esters

- Hydroxyl groups of sugars can be esterified to form acetates, propionates, benzoates, etc
- Sugar phosphates are of great biological importance.
- Metabolism of sugars inside the body starts with phosphorylation.

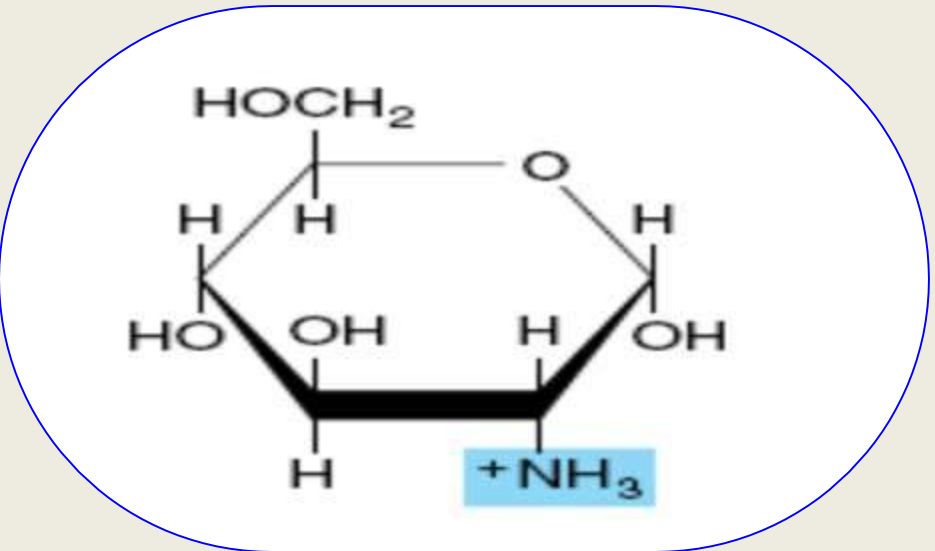
  e.g Glucose 6-$P0_4$

Amino sugars

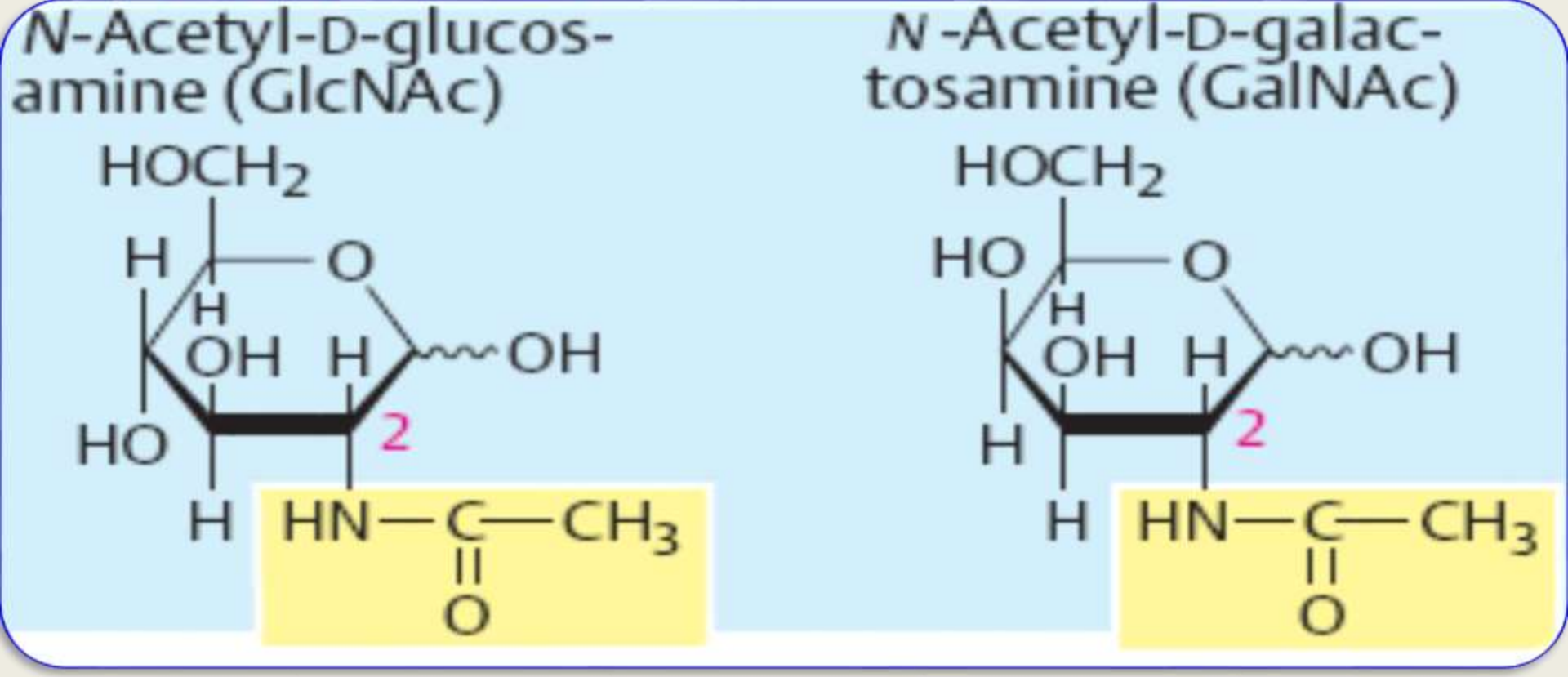
- Amino groups may be substituted for hydroxyl groups of sugars to give rise to amino sugar.
- Generally the group is added to the second C of hexoses.
- They are non –reducing and do not form osazones
- They are found in GAGS, glycoproteins, proteoglycans
- Abbreviations:-

  GluNac = N-acetyl –glucosamine

  GalNac =N-acetyl-galactosamine



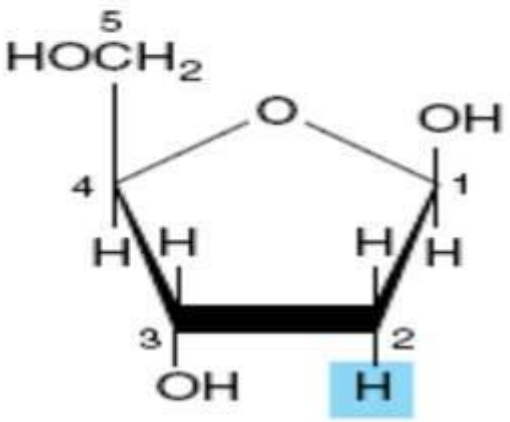
GLUCOSAMINE or 2 amino-D-glucopyranose (α form)



- The amino group may be further acetylated to produce N-acetlyated sugars like N-acetly glucosamine (GlcNac) or N-acetyl galactosamine (GalNac) which are important constituents of glycoproteins and MPS

Deoxy Sugars

- Oxygen of the hydroxyl group may be removed to form deoxy sugars.
- They are non reducing.
- Don’t form osazones.
- Deoxyribose is present in DNA



2-Deoxy-D-ribofuranose (β form).

Disaccharides

- When two monosaccharides are combined together by glycosidic linkage, a disaccharide is formed.
- Two types:-

Non-reducing

Sucrose	Cane sugar
Trehalose	in yeast

Reducing

Lactose	Milk sugar
Maltose	Malt sugar

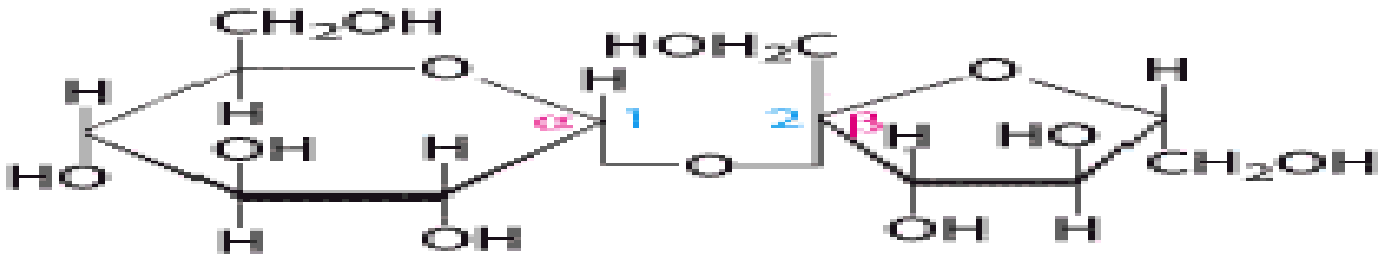
Sucrose

- Cane sugar, table sugar
- Glu + Fru (1→2)
- Sweetening agent
- Non-reducing
- No osazones
- Clinical Importance:-

  -dental caries

  -Bypasses metabolic check points- OBESITY

  -“Sucrase deficiency “



Sucrose

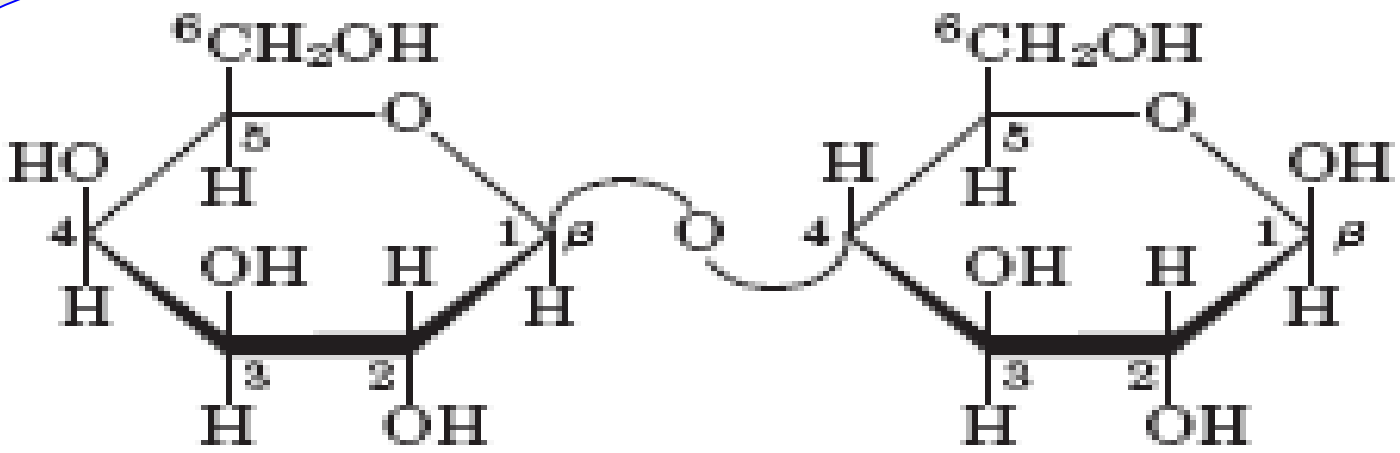
(α-D-Glucopyranosyl-(1 → 2)-β-D-fructofuranose

Inversion

- Hydrolysis of sucrose (optical rotation +66.5^{o}) will produce one molecule of glucose (+52.5^{o}) and one molecule of fructose (-92°)
- Therefore the products will change the dextrorotation to levorotation (INVERSION)
- Equimolecular mixture of glucose and fructose thus formed is called as Invert Sugar
- The enzyme producing hydrolysis of sucrose is called INVERTASE

Lactose

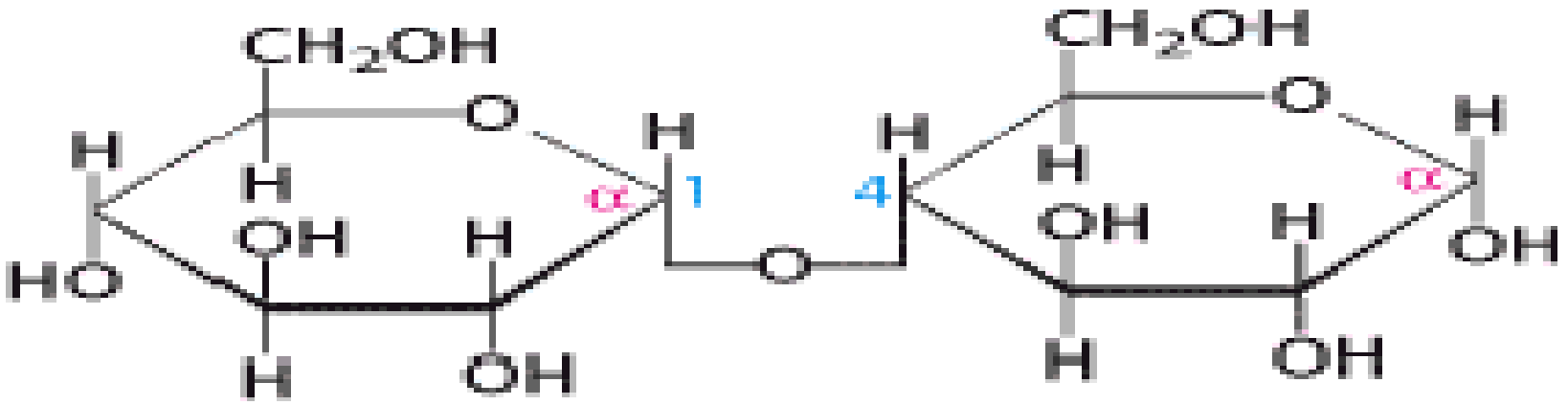
- Milk sugar
- Galactose + Glucose (β1→4)
- Reducing disaccharide
- Beta glycosidic linkage
- Osazone – Powder Puff or hedgehog shaped



Lactose (β form)
β-D-galactopyranosyl-(1→4)-β-D-glucopyranose
Gal(β1→4)Glc

Maltose

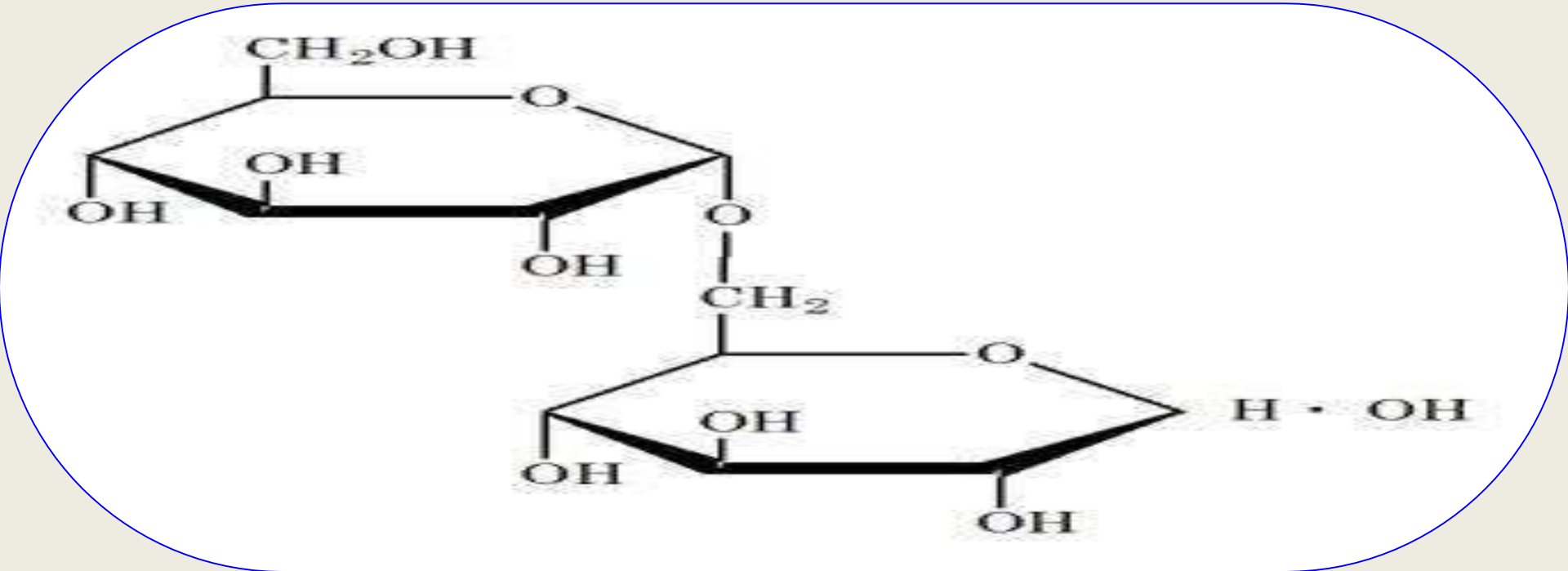
- 2 glucose residues (α1→ 4 linkage)
- Reducing disaccharide
- Malt sugar
- Osazone:- Star shaped or flower petal shaped



Maltose
(α-D-Glucopyranosyl-(1 → 4)-α-D-glucopyranose

Isomaltose

- 2 Glucose in α (1→6) linkage
- Reducing disaccharide
- Produced during partial hydrolysis of starch and glycogen

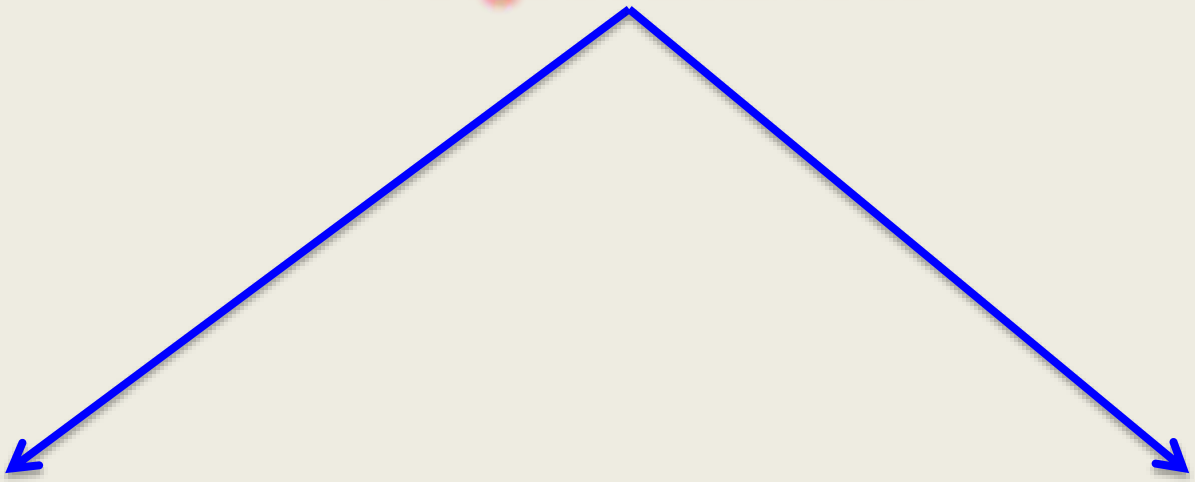


Disaccharides of importance

Sugar	Composition	Source	Clinical Significance
Sucrose	D-glucopyranosyl-(1-2) D-fructo furanoside	Cane and beet sugar, sorghum and some fruits and vegetables	Rare genetic lack of sucrase leads to sucrose intolerance—diarrhea and flatulence
Lactose	D-galactopyranosyl-(1-4)D-glucopyranose	Milk	Lack of lactase leads to lactose intolerance diarrhea and flatulence; may be excreted in the urine in pregnancy

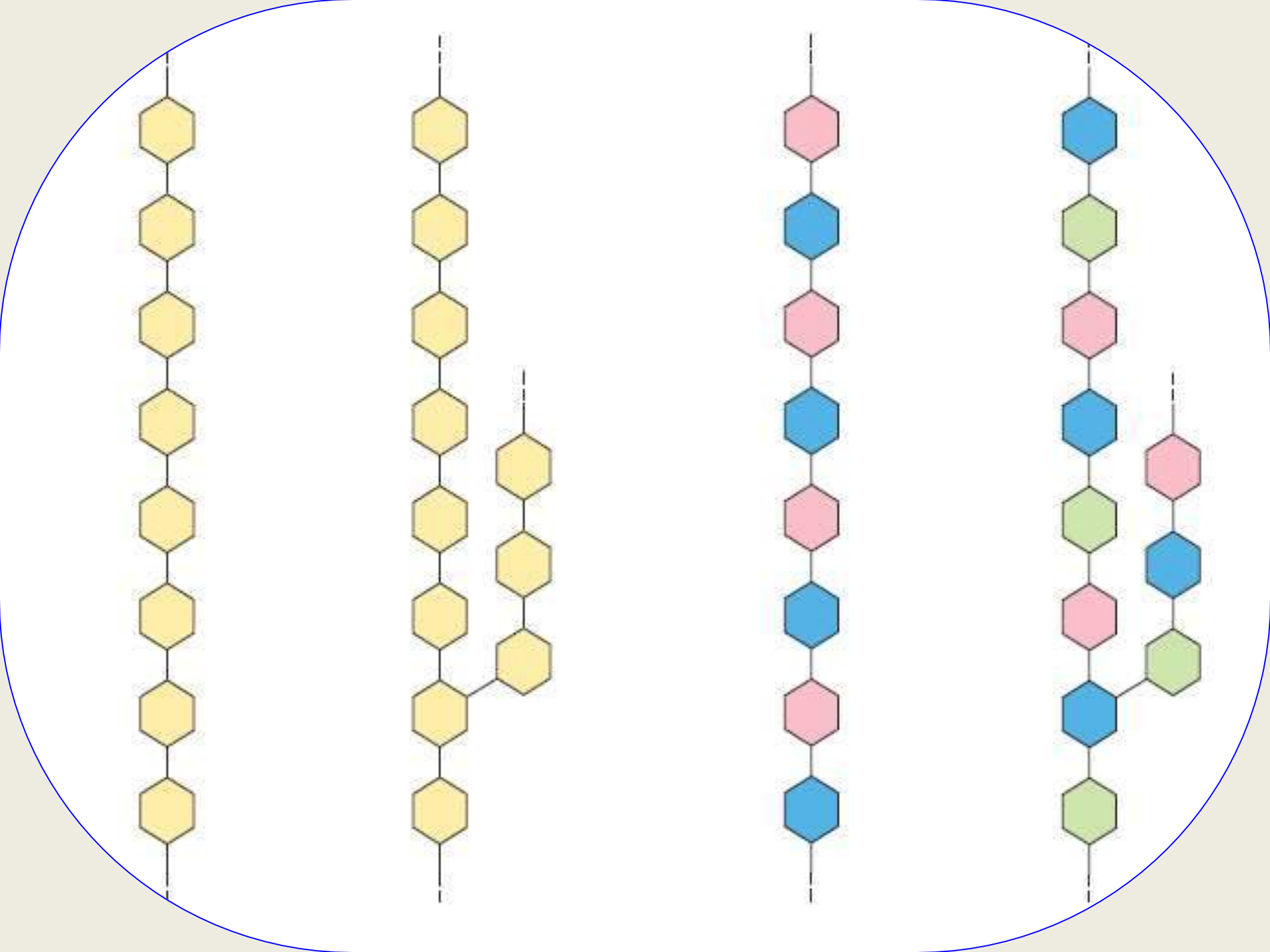
Sugar	Composition	Source	Clinical Significance
Maltose	D-glucopyranosyl-(1-4)-D-glucopyranose	Enzymatic hydrolysis of starch (amylase); germinating cereals and malt	
Isomaltose	D-glucopyranosyl-(1-6)-D-glucopyranose	Enzymatic hydrolysis of starch (the branch points in amylopectin)	
Lactulose	D-galactopyranosyl-(1-4)-D-fructofuranose	Heated milk (small amounts), mainly synthetic	Not hydrolyzed by intestinal enzymes, but fermented by intestinal bacteria; used as a mild osmotic laxative
Trehalose	D-glucopyranosyl-(1-1)-D-glucopyranoside	Yeasts and fungi; the main sugar of insect hemolymph	

Polysaccharides



Homoglycan
Or
Homopolysaccharide

Heteroglycan
Or
Heteropolysaccharide



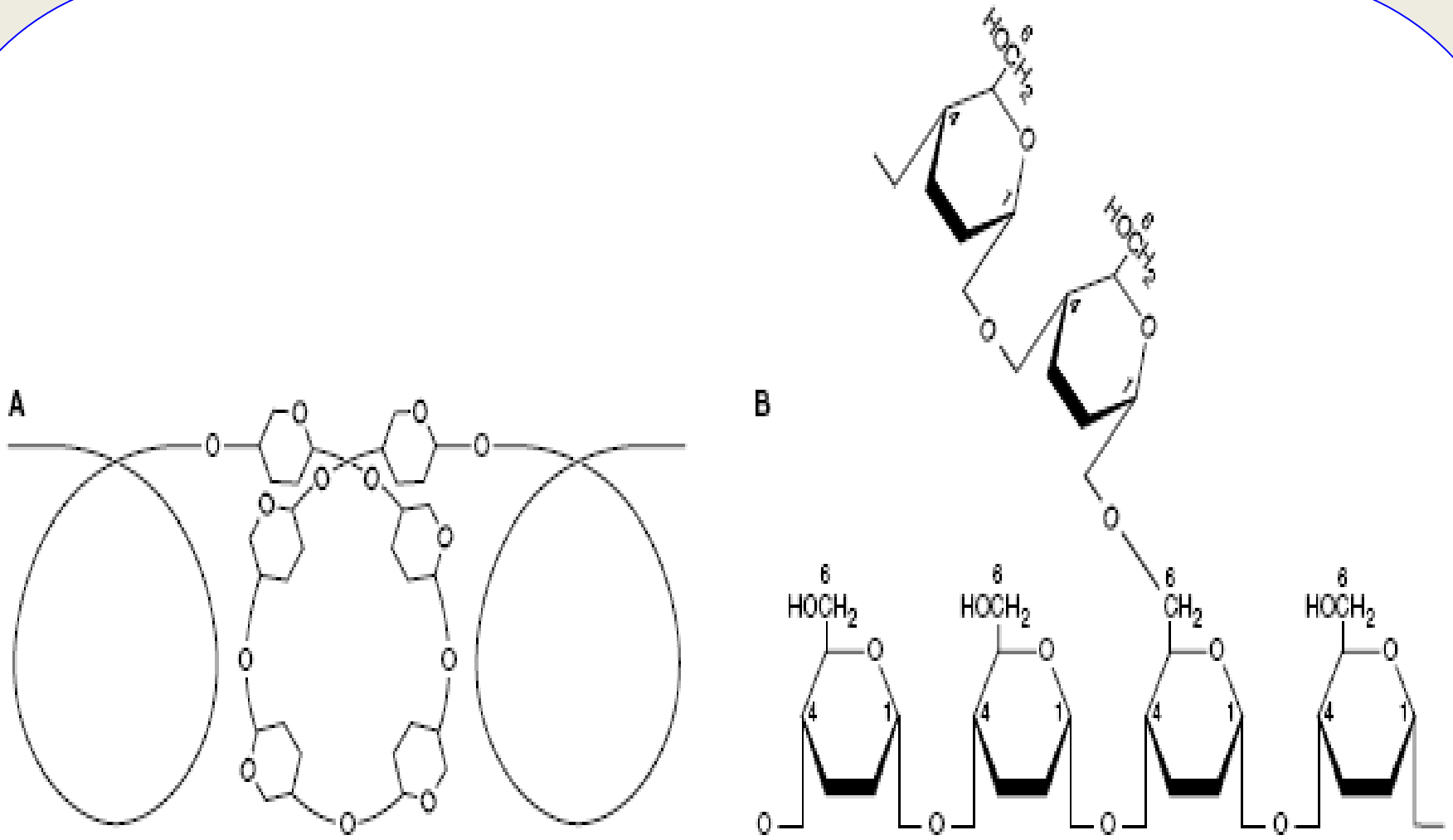
Homoglycans

- Starch
- Glycogen
- Cellulose
- Inulin
- Dextrans
- Chitin

Starch

- Carbohydrates of the plant kingdom
- Sources:
- Potatoes, tapioca, cereals (rice, wheat) and other food grains
- Composed of Amylose & Amylopectin
- **Amylose:**
- When starch is treated with boiling water, 10 -20 % is solubilized.
- This part is called amylose, contains glucose units with α-1,4 glycosidic linkages.
- Mol wt =400,000 or more

- **Amylopectin:**
- The insoluble part absorbs water and forms paste like gel;
- This is called as amylopectin.
- Amylopectin is also made up of glucose units, but is highly branched with molecular weight more than 1 million.
- The branching points are made by α- 1, 6 linkage


A
B
HOCH2
CH2
O
1
4
6

Hydrolysis of starch

- Starch will form a blue coloured complex with iodine; this color disappears on heating and reappears when cooled.
- This is a sensitive test for starch.
- When starch is hydrolyzed by mild acid, smaller and smaller fragments are produced.
- The hydrolysis for a short time produces amylodextrin (violet color with iodine and non-reducing).
- Further hydrolysis…….

amylodex →erythrodex→archrodextrin→Maltose

Violet	Red	no color	no color
Non reducing	Non reducing	Reducing	Reducing

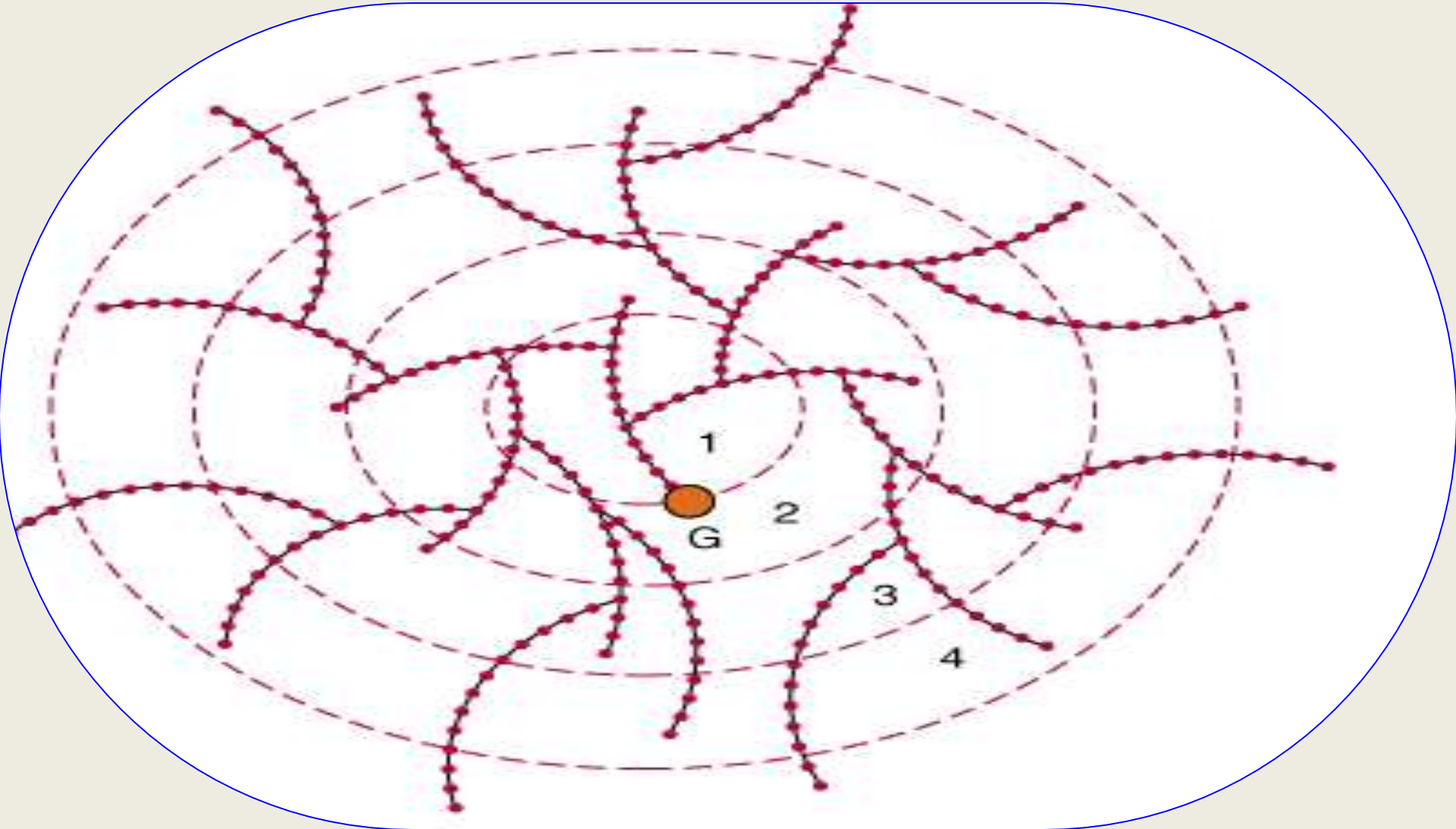
Action of amylases on starch

- Salivary amylases and pancreatic amylases are α amylases, which act randomly on α, 1-4 linkages to split starch into smaller units called dextrins
- Beta amylases (plant origin – almonds etc) act consecutively from one end.
- When beta amylases reach a branch point in amylopectin, enzyme is blocked, leaving a large molecule called as LIMIT DEXTRIN

Glycogen

- Storage form of energy in animal.
- Stored in liver and muscle
- Stores more glucose residues per gram than starch.
- More branched and compact than starch.
- Less osmotic pressure.
- More energy in a smaller space.
- Glycogen in liver (6-8%) is higher than that in the muscles (1-2%).
- Liver glycogen - first line of defense against declining blood glucose levels especially between meals.

- A homopolysaccharide: linear chain of $\alpha(1\rightarrow4)$ linked glucosyl residues with branches joined by α $(1\rightarrow6)$ linkages



Cellulose

- Glucose units combined by β-1,4 linkages.
- Straight line str. with no branches.
- Mol wt 2-5 million.
- This bond is digested by cellobiases an enzyme not present in humans.
- *Herbivores animals have large caecum which harbor bacteria which break cellulose.*
- White ants (termites) and some wood fungi also have cellulase.
- Commercial applications: nitrocellulose, cellulose acetate membranes for electrophoresis ETC

Inulin

- D -fructose in β-1,2 linkages.
- Source:
- Bulbs and tubers chicory, dahlia, dandelion, onions, garlic.
- Not metabolized .
- Not absorbed nor secreted by kidneys.
- USE – to measure GFR.

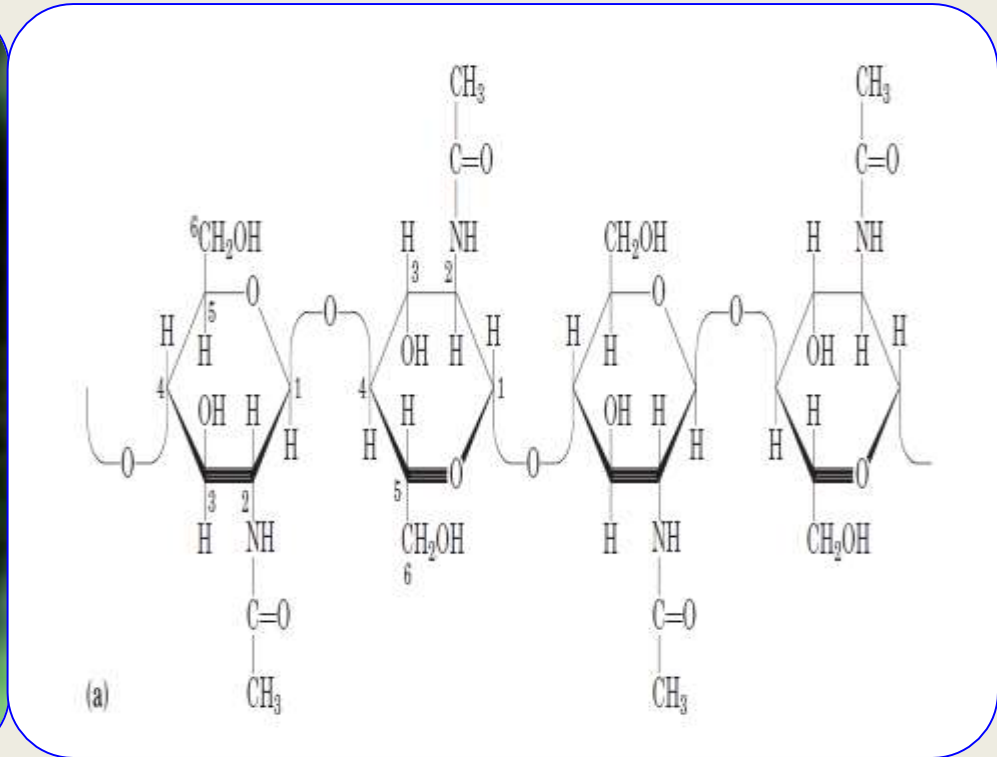
Dextrans

- Highly branched homoglycan containing Glu residues in 1-6, 1-4 and 1-3 linkages.
- Produced by microbes.
- Mol. wt:- 1-4 million.
- As large sized, they will not move out of vascular compartment so used as plasma expanders.

Chitin

- N-acetyl glucosamine with beta 1,4 glycosidic linkage
- Exoskeleton of crustacea and insects.





Heteroglycans

Agar and Agarose

- It is made up of D-galactose and an L-galactose derivative ether – linked between C-3 and C-6
- It is dissolved in water at 100 ^{o} C, which upon cooling sets into a gel.
- Agar cannot be digested by bacteria so it is widely used as a supporting media to culture bacterial colonies.
- Also used as a supporting agent for immuno-diffusion and immuno-electrophoresis.
- Two components : Agarose (unbranched)

  Agaropectin (branched)
- Agarose is made up of D- galactose combined with 3,6-anhydro L-galactose units and is used as a matrix for electrophoresis.

Mucopolysaccharides or GAG

[URONIC ACID + AMINO SUGAR] $_n$

Acetylated amino sugars, sulfate and carboxyl groups may be present also

Heteropolysaccharides

- Polymers made from more than one kind of monosaccharides or monosaccharide derivatives.
- Eg : Glycosoaminoglycans,
  Agar
  Agarose

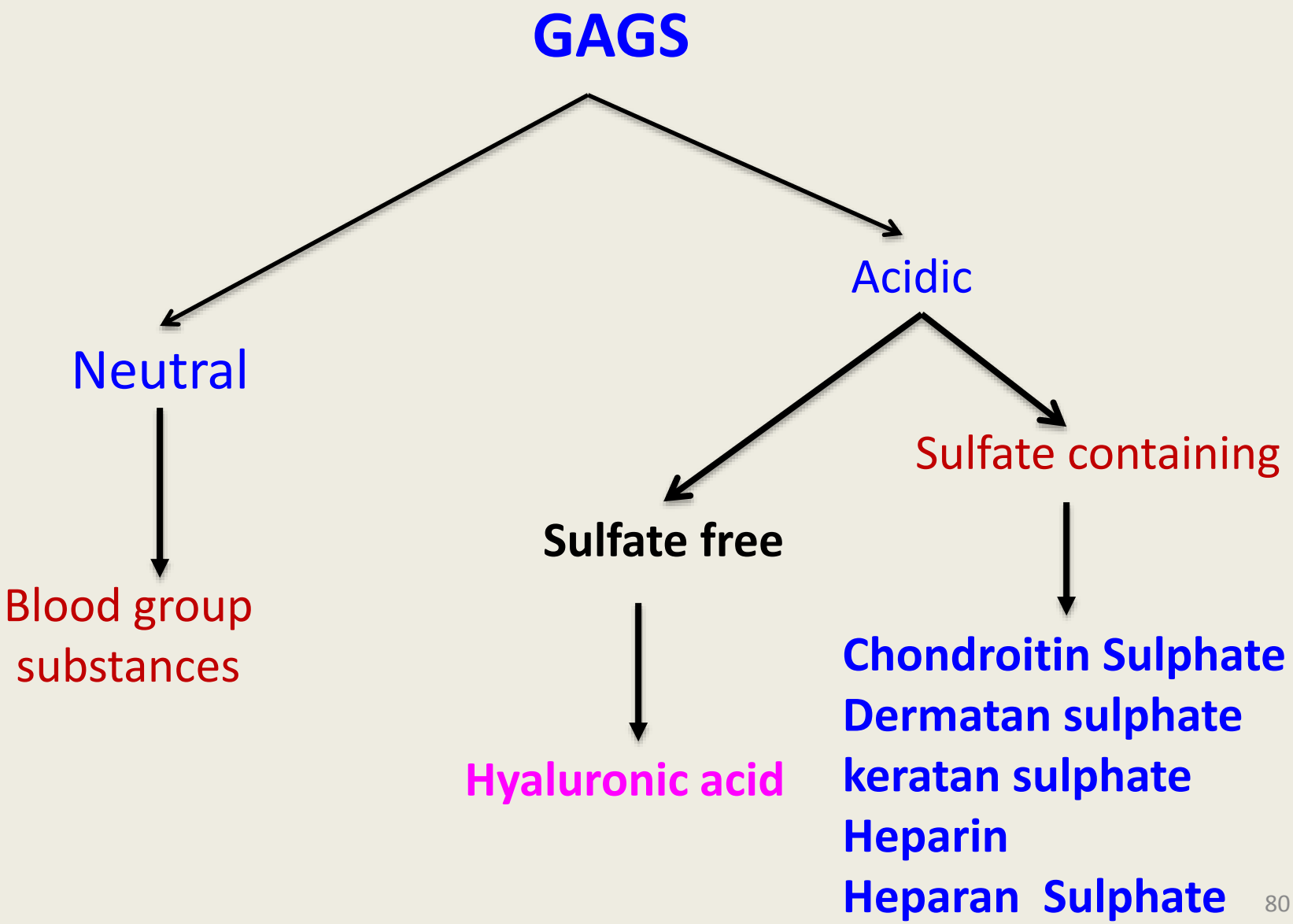
- Long, Unbranched heteropolysaccharide, made of repeating disaccharide units containing uronic acid & amino sugars.
- Amino sugar – Glucosamine or Galactosamine
  (Present in there acetylated form)
- Uronic acid – D-Glucuronic acid or L-Iduronic acid
- GAGs are the most important group of heteroglycan in humans.

- First isolated from mucin so called mucopolysaccharides.
- Major components of extracellular matrix of connective tissue, including bone and cartilage, synovial fluid, vitreous humor and secretions of mucus producing cells.

- Gel forming component of extracellular matrix
- The anionic groups (carboxy & sulfate groups) being strongly hydrophilic tend to bind large amount of water producing gel like matrix, that forms the bodies ground substance.
- Heteropolysaccharide chains repel one another and therefore exist in extended conformation in solutions.
- This produces slippery consistency of mucus secretions and synovial fluid.

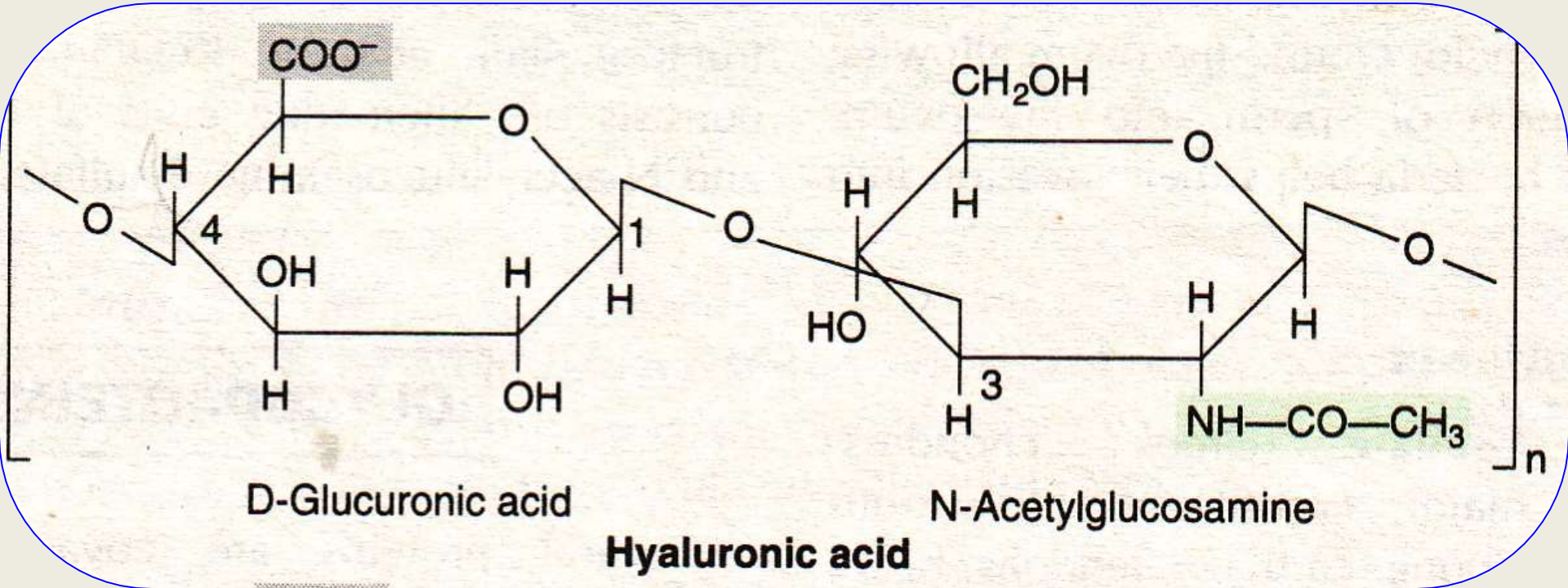
- Structural support to connective tissue
- GAGs form matrix or ground substance that stabilizes and supports the cellular and fibrous components of tissues.
- Other functions:
- Plays an important role in mediating cell-cell interactions
- Their slippery consistency makes them suitable for a lubricant action in joints.

Classification



Hyaluronic acid

- It is sulfate free GAG.
- Synovial fluid of joints, vitreous humor, connective tissues and cartilage.



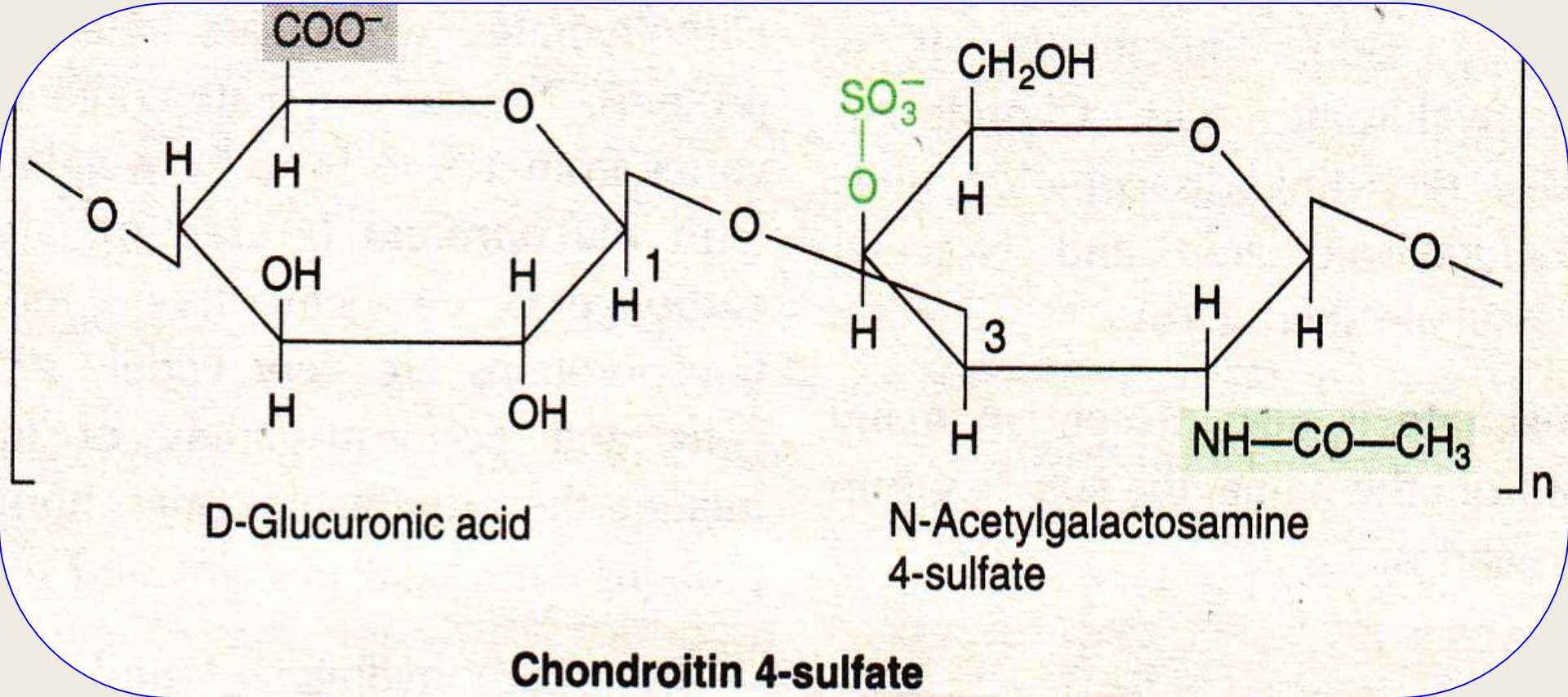
Functions of Hyaluronic acid

- Serves as a lubricant and shock absorbant in joints.
- Acts as seives in extracellular matrix.
- Permits cell migration during morphogenesis & wound repair.
- **Hyaluronidase** is an enzyme that breaks β1 – 4 linkages of hyaluronic acid.
- Present in high concentration in seminal fluid, & in certain snake and insect venoms.

- Hyaluronidase enzyme of semen degrades the gel around ovum & allows effective penetration of sperm into ovum, thus helps in fertilization.
- The invasive power of some pathogenic organism may be increased because they secrete hyaluronidase.

Chondroitin 4-sulfate

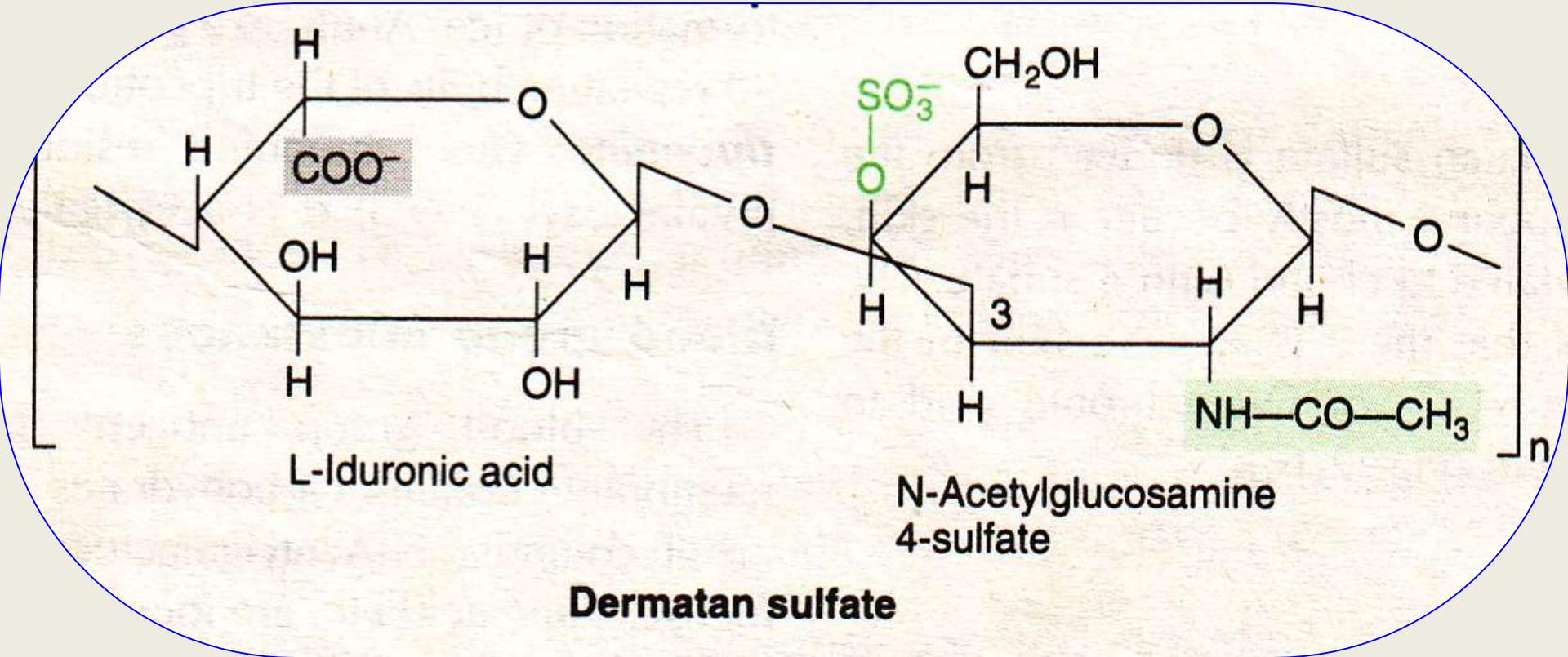
- Most abundant GAG in body.



- Widely distributed in bone, cartilage & tendons.
- Functions:
- In cartilage, it binds collagen & hold fibers in a tight strong network.
- Role in Compressibility of cartilage in weight bearing.

Dermatan sulfate

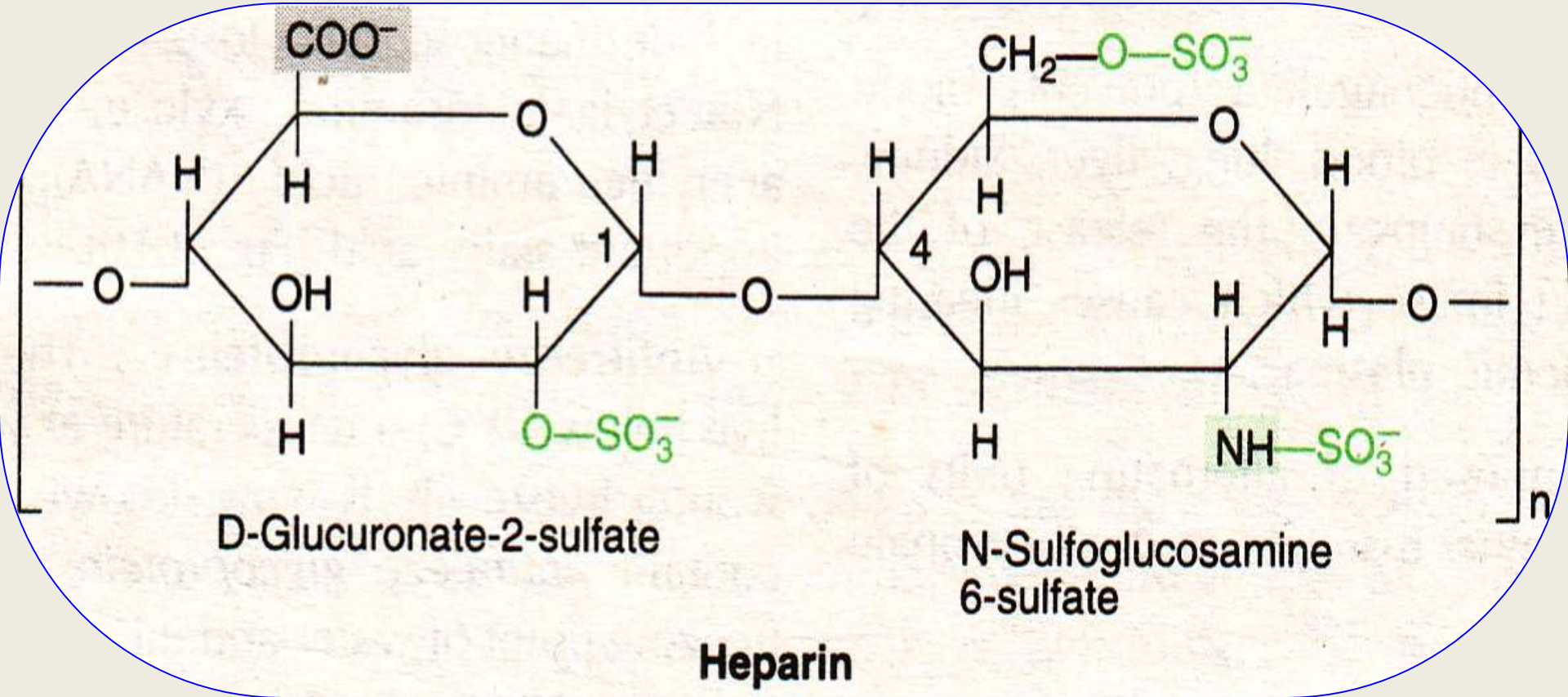
- Contains repeating units of L-iduronic acid and N-acetyl glucosamine 4 sulfate.



- Present in skin, cardiac valves & tendon.
- Function:
- Present in sclera of eye where it has important function in maintaining overall shape of eye.

Heparin

- Only GAG present intracellular: In granules of mast cells and also in lung, liver and skin.



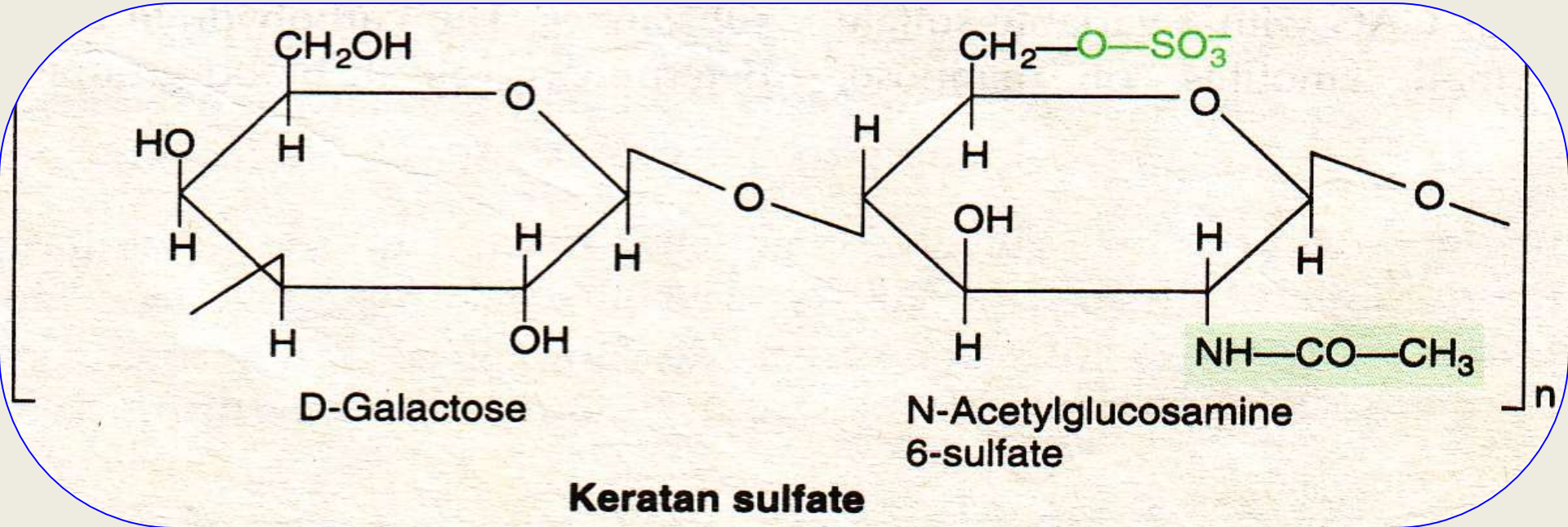
- Strongly acidic due to presence of more sulfate group.
- Functions:
- It is an anticoagulant (prevents blood clotting)
- Heparin helps in the release of the enzyme lipoprotein lipase (LPL) which helps to clear the lipidemia after fatty meal – so called clearing factor.

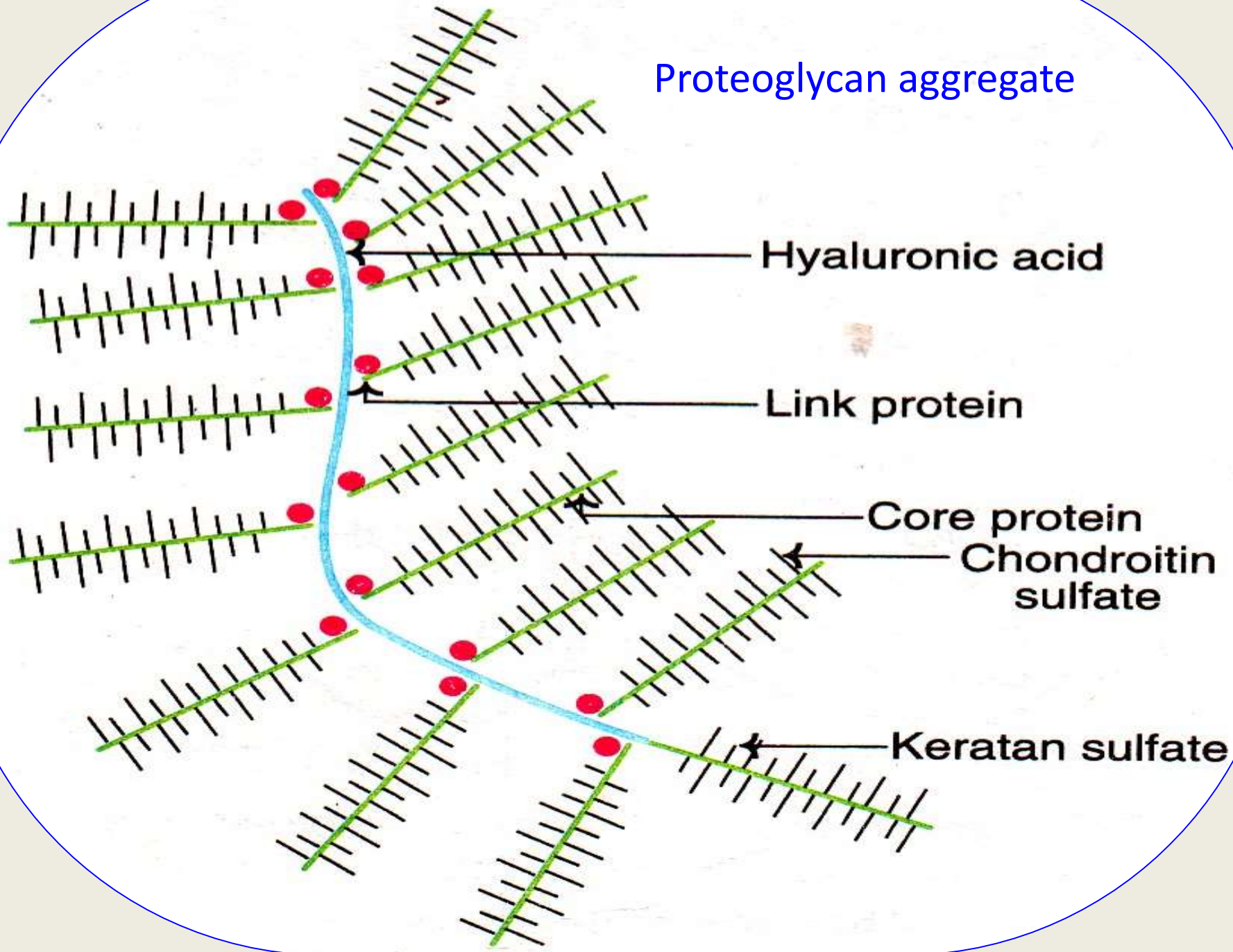
Heparan sulfate

- Structurally similar to heparin, but has a
- Lower molecular weight
- Contains higher acetyl groups & less sulfate group
- Predominant uronic acid is D-Glucuronic acid
- It is an extracellular GAG found in basement membrane and is an essential component of cell surfaces.
- Determines charge selectiveness of renal glomerulus.

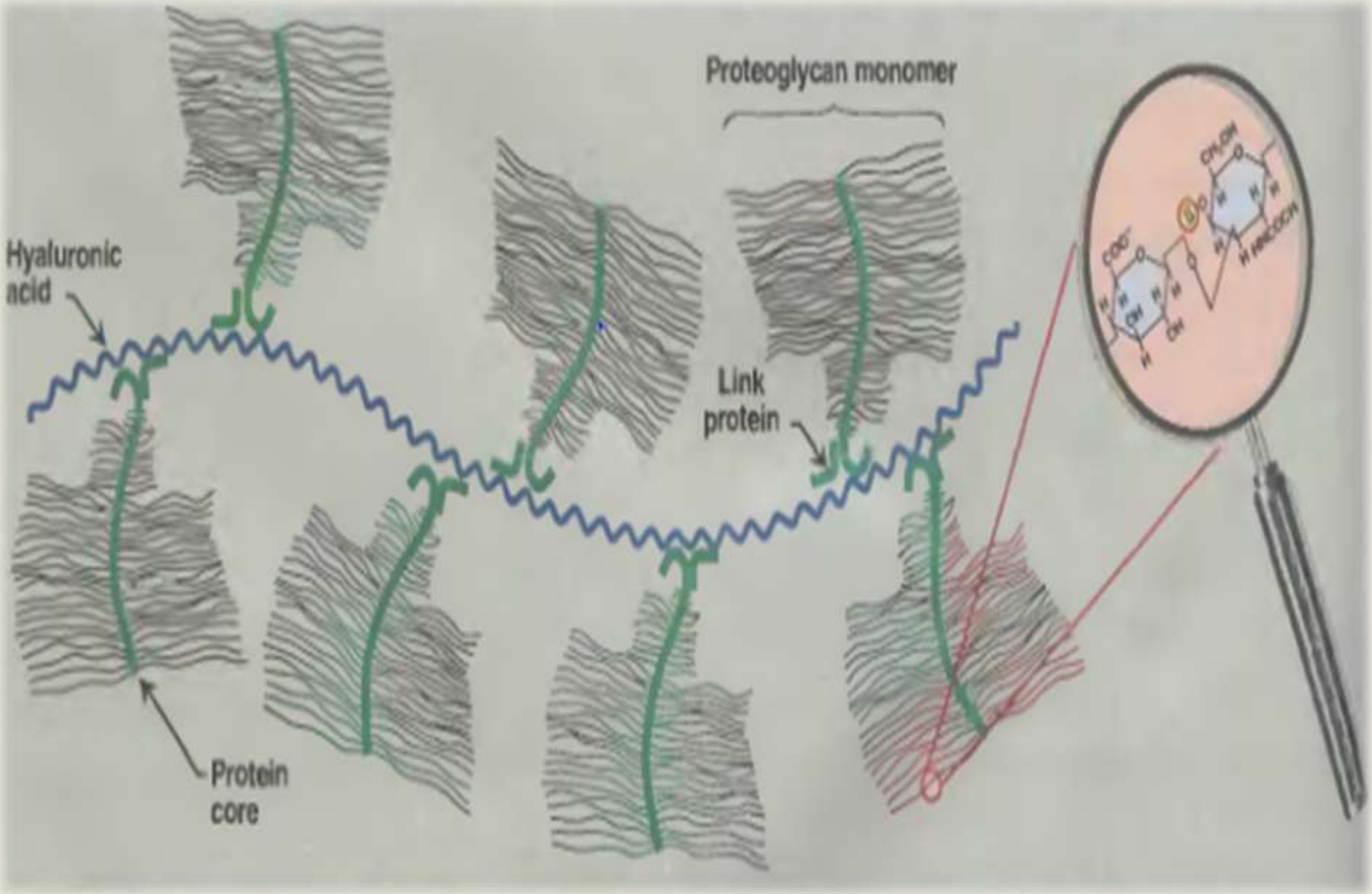
Keratan sulfate

- Only GAG with no uronic acid.
- Found in cornea & tendons.
- Function:
- Maintains the corneal transparency.




Proteoglycan aggregate
Hyaluronic acid
Link protein
Core protein
Chondroitin sulfate
Keratan sulfate

Proteoglycan aggregate



Blood group substances (blood gr Antigens)

- RBC membrane contains several antigenic substance, based on which classified into different blood groups.
- They contain carbohydrates as glycoproteins or glycolipids.
- N-Acetylgalactosamine, galactose, fucose, sialic acid etc are found in blood gr substances.
- Carbohydrate content plays a determinant role in blood grouping.

Agar

- Contains galactose, glucose & other sugars.
- Obtained from sea weeds
- Functions:
- Cannot be digested by bacteria.
- So used as supporting agent to culture bacterial colonies.
- Also as support medium of immuno diffusion & immuno-electrophoresis.

Agarose

- Galactose and 3,6 anhydrous galactose units
- Used as matrix for electrophoresis.

Proteoglycan	Glycoprotein
GAGs are covalently attached to a protein (95 % sugar)	Primarily of protein with small amount of carbohydrates
Sugar chain are longer (100 or more sugars)	Short oligosaccharide chain (>20)
Linear	Highly Branched
Disaccharide repeats	Do not have repeat sequence

Reference Books

- Test Book of Biochemistry- Harper

Lippincott's Illustrated Reviews

- Test Book of Medical Biochemistry – MN Chatterjea

Thank you